

Introduction to Photography with the Leica M EV1

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To Lucie, Maël, Margot, Maxime, and Sacha

Forward

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Introduction to Photography with the Leica M EV1

Patrick Cousot

Abstract

A short, simple, and illustrated introduction to the fundamental concepts of photography (with a few optional technical explanations), and their practical application with a Leica M EV1.

1 The Leica M EV1 Camera

Leica is a camera maker company that brought to prominence the 35mm film camera in 1925 (Leica A), the rangefinder camera in 1932 (Leica II), the M-mount lenses in 1954 (Leica M3) and their digital versions since 2006 (Leica M8). The Leica M11 first appeared in 2022.

The Leica M EV1 was released in 2025. It has a viewfinder for manual focussing and composition/framing, but no rangefinder. So the Leica M EV1 is the first Leica M without the traditional Leica optical and mechanical rangefinder. Its use is similar to that of a Leica M11 with an external electronic viewfinder Visoflex 2 replicating the screen together with focus peaking. The electronic viewfinder is integrated

within the Leica M EV1 camera and offers a better image quality (5.76 mega pixels (MP) instead of 3.68 MP display). The Leica M EV1 is also smaller and lighter than the M11.



Figure 1: Leica M-EV1 (here with a thumb support, a soft release button, a carry strap, and a [M-mount](#) SUMMILUX-M 1:1.4/35 ASPH lens with UV filter (see section [41](#)) and screwed lens hood).

Always carry the camera while holding the strap to avoid falls.

We explain in simple terms how to use the Leica M EV1 (clicking on dark blue text refers to another section of the e-booklet or to [Wikipedia](#) for more detailed and scientific explanations).

2 Preparation of the Leica M EV1

Before taking a [photo](#), the Leica M EV1 must be prepared by inserting a [SD card](#) (Secure Digital card), a [battery](#), initializing the SD card, and mounting a lens.

2.1 Inserting a SD Card and the Battery



- Make sure the camera is off,



by turning the main switch anti-clockwise;

- Unload the battery by turning the battery release lever, so that the battery pushes out slightly, then push down the battery with a brief impulse, and then pull the battery out. These two successive actions prevent an inadvertent fall of the battery.



- Insert an new blank SD card;
- Orient the SD card so as to see the contacts, as shown on the camera battery slot;



- Push the SD card, visible contacts down, into its reader until it clicks.



To unload the SD card, slightly and quickly push the SD card down, and then pull the card out. Again, these two movements required to remove the card prevent an inadvertent fall of the card.

- Charge the battery (Leica BP-SCL7 Lithium-Ion Battery).



which may take 3.5 hours for a full charge, reaching 80% in about 2 hours (when the orange indicator lights up).

- Insert the charged battery by pushing the battery into its compartment until it clicks.

2.2 Initialize your SD card (the first time only)

- Turn the camera on,



by turning the main switch clockwise;

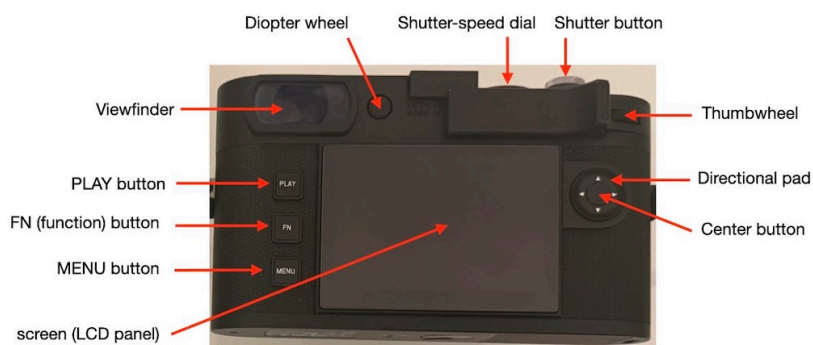
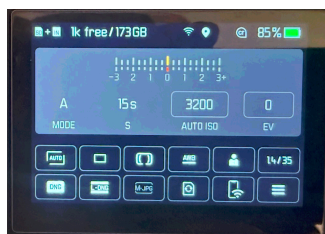
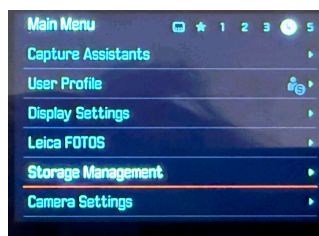


Figure 2: Back of the Leica M EV1

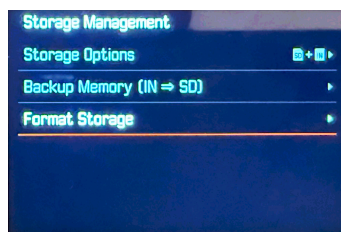
- Look at the back of the camera (see figure 2) and press the PLAY button  to make sure there are no photos on the card. Press PLAY again to exit the photo review mode.
- Press the MENU button  to obtain the following screen:



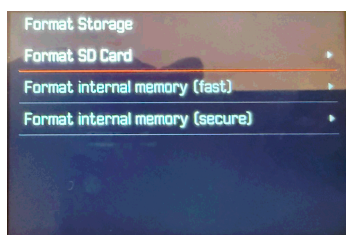
- Press again five times the MENU button to select the fourth submenu (marked ④) and then press 4 times the down arrow ▾ of the directional pad  to get



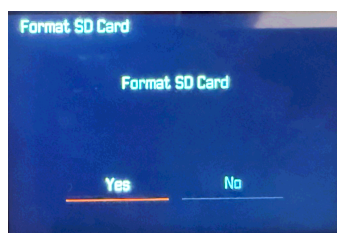
- Press the Center button to select Storage Management and then press the down arrow ▾ of the directional pad  to get:



- Press the Center button to select Format storage and get



- Push the Center button and then the left arrow to select Yes;



- Pressing the Center button will start the formatting of the SD card.



This takes a few seconds and at the end returns to the Format SD Card menu.

- Shut down the camera



2.3 Menus

The previous sequence of actions to navigate the [menus](#) of the M EV1 is traditionally denoted as “ON → MENU → ④ → Storage Management → Format Storage → Format SD Card → Yes → OFF”. The selection of the next step by pressing arrows Δ , $\triangleright$, ∇ , $\triangleleft$, the central button, or MENU (to directly get from ① to ④ instead of scrolling down through several pages with ∇) is left implicit. ON, MENU, and OFF are also omitted for brevity. Pressing MENU several consecutive times helps to quickly navigate to the menu pages numbered ① to ⑤.

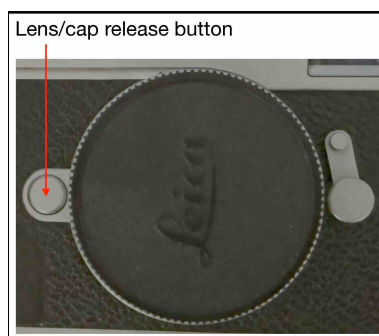
2.4 Mounting a lens

The [Leica M-mount](#) allows many different [lenses](#) to be mounted on the M EV1 camera body.

- First remove the rear cap of the lens by turning it anti-clockwise (here a NOCTILUX-M 1:0.95/50 ASPH) to see the [M-mount](#).



- Then remove the [camera cap \(lens cover\)](#) by maintaining the lens/cap release button pressed down and then turning the cap left.



- Then mount the lens (a SUMMARON-M 1:5.6/28mm in the example)

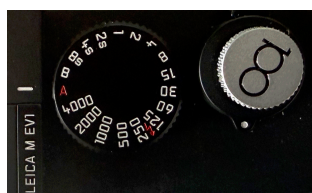


(by aligning the two red buttons on the camera and the lens and then turning the lens clockwise to the right until hearing a click, which is important to make sure that the lens is correctly fixed on the camera).

To unmount the lens, maintain the lens release button in the front of the camera pushed down and then turn the lens anti-clockwise to the left. Put back the caps on the lens and the camera.

3 The First Photo With a Leica M EV1

- Set the ISO sensitivity to Auto ISO (automatic), with MENU → ② → ISO → Auto ISO;
- Turn the shutter speed to **A** (automatic);



Shutter speed dial

- Turn the aperture on the lens to its maximum (16 or 22), and turn the distance on the lens to ∞ ;



- Turn the camera on,



by turning the main switch clockwise.

- Get the lens cap off, if any.
- Look to a [landscape](#) or a distant subject on the screen (this may require to first press the shutter button half-way if the camera got in standby mode).



The red lines show objects in [focus](#) (see section [21.2.3](#)). The camera settings will remain the same as long as the shutter button is pressed halfway. Release and press halfway again if another view is preferred.

- Instead of looking to the screen you can have the same image and focussing information with the viewfinder. The diopter wheel on [figure 2](#) can be used to adjust the viewfinder to your view if you don't want to shoot with glasses [[23](#), diopter setting, page [32](#)].
- When satisfied with the screen view, press the shutter button down fully to take the photo. A very recognizable sound produced by the [mechanical shutter](#) should be audible.



- Look at your photo on the camera screen.

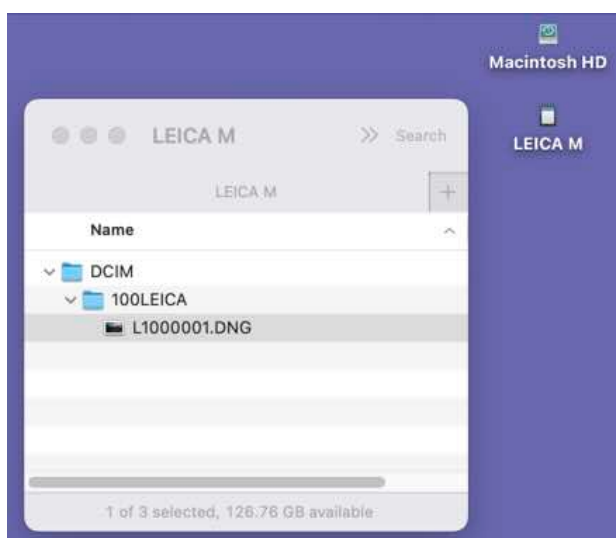


The screen will go black after 30s, push the shutter button halfway to reactivate.

- Transfer the photo
 - to a computer by directly connecting the M EV1 camera to the computer with a [USB-C](#) cable.



- or, to a computer by removing the battery, then ejecting the SD card, and finally inserting it in your computer slot or a [SD card reader](#).



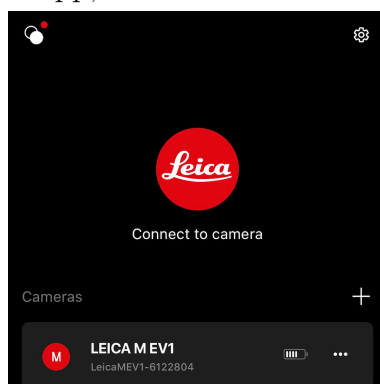
Opening the file [L1000001.DNG](#) with [Preview](#) or [Graphic Converter](#) on [MacOS](#) will show the photo enlarged on the computer screen. It can be saved as [PDF format](#) and then opened with [Adobe Acrobat Reader](#).

- to an [iPhone](#) using the [Leica FOTOS app](#).
- First pair the [iPhone](#) with the camera by opening the Leica FOTOS app on the [iPhone](#),

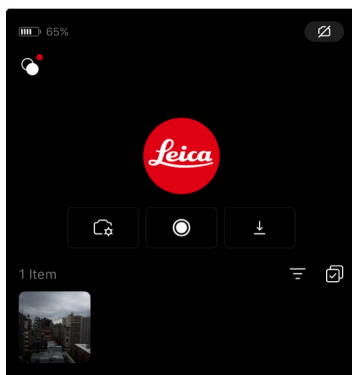


selecting **Pairing** in the Leica FOTOS camera menu, and following the instructions (you might have to re-activate the camera by pushing the shutter button halfway).

- Once the [iPhone](#) and camera are paired, open the Leica FOTOS app,



click on LEICA M EV1 to connect to the camera (which must be on and active, half-press the shutter button if not) to get the photo

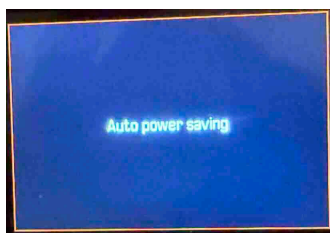


click (or double-click) on the photo to enlarge it and  to send it electronically (which may require to turn the camera off to allow the [iPhone](#) to chose the destination).

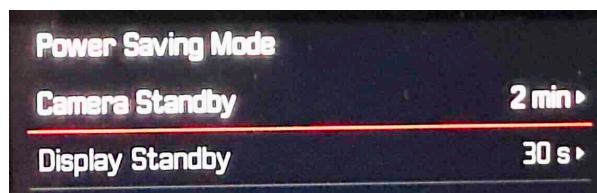
Read what follows to understand what you need to know for your next photos!

4 Silent Deactivation of the Camera and Reactivation

When the camera is on, it automatically goes to inactive mode with a dark screen after some time. This saves energy for the battery. Pressing the shutter button halfway or switching the main switch off and on again will reactivate the camera, as shown by a red Status [LED \(light-emitting diode\)](#) and a visible screen. The silent deactivation time



is 2 min by default and can be chosen by the photographer with MENU → ④ → Camera Settings → Power Saving Modes → Camera Standby to get



Then the desired Camera Standby time can be turned Off or chosen between 30 s and 60 min.

Similarly the display deactivates after 30 s, which can also be modified by MENU → ④ → Camera Settings → Power Saving Modes → Display Standby to be Off or between 30s and 5 min.

Pressing the shutter button halfway or switching the main switch off and on again will reactivate the camera and screen.

5 Light

Photography is the art of fixing **visible light** on a support. We see and the camera records light from a source (sun, artificial lightning) that **reflects** on objects.

Light propagates in [straight line beams](#) (think to the straight rays of the sun visible through a cloud), of course except for [reflexion](#) on a mirror and [refraction](#), for example through a liquid.

Light has an [intensity](#) giving an impression of [brightness](#) or [brilliance](#) (in absence of clouds, the light of the moon is less intense than the light of the sun at sunrise or sunset, which is itself less intense than the light of the sun at noon).

Light is also a [quantity](#) (the quantity of light received from the moon for a long time is the same as the quantity of light received from the sun during a very brief instant).

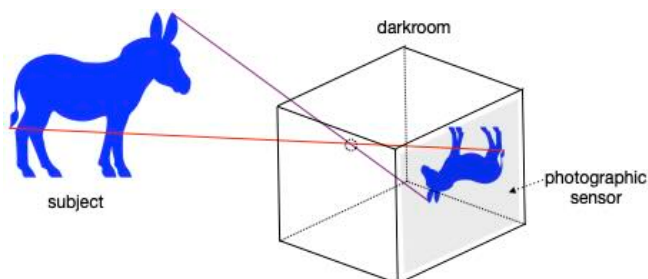
Light can be [polarized](#) by reflection on metal or water which alter how the light is transmitted and seen (for example polarization of light allows the glare-reducing effect of polarized sunglasses. [Polarizers](#) in photography annihilate light polarization, see section 45).

More scientific details on en.wikipedia.org/wiki/Light, [en.wikipedia.org/wiki/Polarization_\(waves\)](http://en.wikipedia.org/wiki/Polarization_(waves)), or [39].

6 Darkroom

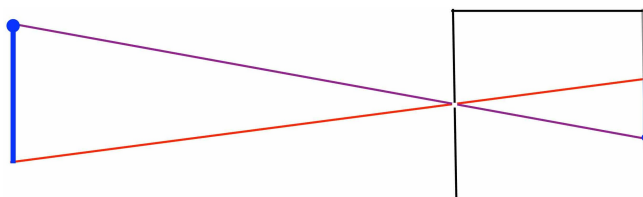
The [darkroom](#) has been known for centuries and is at the origin of photography. A darkroom is a box with a small pinhole in one face and a image sensor on the opposite face. This sensor was originally made of [frosted glass](#) or thin paper. The [light beams](#) going through the tiny pinhole reach the image sensor. Of course if the light outside the box is intense one sees nothing on the image sensor. But if the image sensor is seen in the dark (by looking at the image sensor while covered by a black cover), one can see an inverted image of the subject (flipped horizontally and vertically, that

is, turned 180°).



This is because a ray of light from the top left of the subject goes through the pinhole and arrives at the bottom right of the image sensor. The same way, a ray of light from the bottom right of the subject goes through the pinhole and arrives at the top left of the image sensor. Of course cameras flip the image to see it right side up on the viewfinder and screen.

Instead of reasoning in three dimensions, one can use a simpler representation of the darkroom in two dimensions, valid in the two horizontal and vertical planes, as follows.



Originally artists painted directly over the frosted glass or thin paper serving as image sensor to reproduce the subject. Nowadays some artists like [Arnulf Rainer](#) and [Philippe Cognée](#), use a similar idea and paint over photographs.

Here is an improvised darkroom made of a cardboard carton with a pinhole in it and a image sensor made of plant-based plastic frosted with a mirror fine [sandpaper](#) of [grit size](#) 1000. The photo of the sensor under a black sheet shows that the luminous cone is inverted.



The [history of photography](#) [42] is a long search of (sometimes dangerous) chemicals able to capture the light on the image sensor of the darkroom and then fix it using a [developer](#) and then a [fixer](#), to be able to see the photo in plain light without further modifications by exposure of the photo to light. [Film photography](#) appeared in the 1900's while digital photography appeared in the 1990's, replacing [film](#) by a [digital image sensor](#).

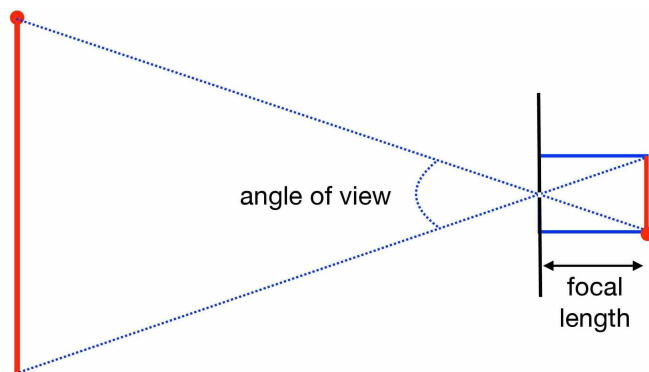
7 Lenses of Different Focal Lengths

The darkroom also allows us to understand why a camera like the M EV1 has several [lenses \(objectives\)](#) of different “[focal lengths](#)” 18 mm ([millimeters](#)), 21 mm, 24 mm, 28 mm, 35 mm, 50 mm, 75 mm, 90 mm, and 135 mm, some with

adjustable focal lengths 16-18-21 mm and the older 28-35-50 mm. Some vintage focal lenses such as the Leica Summicron-C 40mm, the [Konica](#) (M-Hexagon 21-35, 28, 90 mm lenses), or the french Boyer f. Leica M 2,8/45mm Topaz, are no longer produced (the [Établissements Boyer](#) founded in 1895 disappeared in the '70s). Other companies also produce M-mount lenses such as [7Artisans](#) (28, 35, 75 mm), [Lomography](#) (17mm), [Meyer-Optik Görlitz](#) (58 mm), [Voigtländer](#) (10 mm, 15 mm, 40mm), [TTArtisan](#) (35, 50, 100 mm), and [Zeiss](#) (15, 21, 25, 28, 35, 50 mm).

8 Definition of the Focal Length

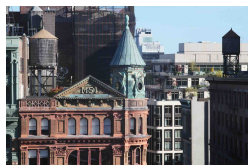
The focal length is the depth of the darkroom. It determines the angle of view and the magnification of a lens.



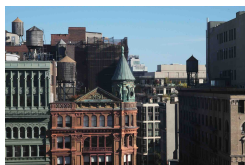
The larger the focal length is, the smaller the (horizontal and vertical) angle of view and the larger the magnification are.

This is clear on the following pictures all taken from the same shooting point and centered on the copper-covered

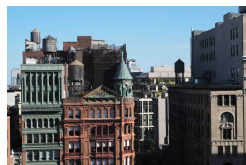
(green) cupola of the former MSI (Manhattan Savings Institution) building in New York, now called the Bleecker Tower.



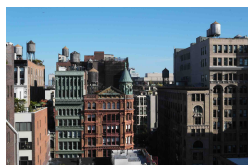
135mm



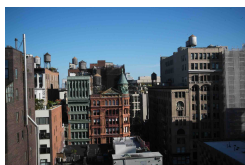
90mm



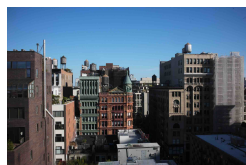
75mm



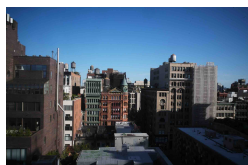
50mm



40mm



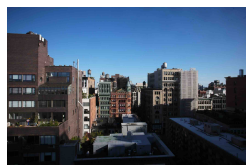
35mm



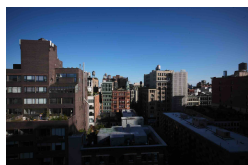
28mm



24mm



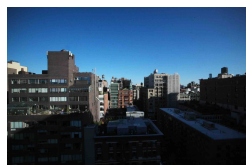
21mm



18mm



16mm

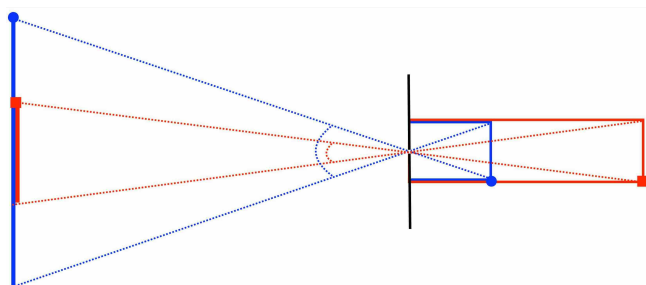


15mm

Observe the water tank (in the background on top right of the picture) appearing proportionally smaller than the cupola with small focal lengths. Let us explain why.

9 Comparing Lenses with Different Focal Lengths

Different focal lengths correspond to different depths of the darkroom box.



The two darkrooms, the small in blue and the large in red, have exactly the same image sensor size in their back but different depths, that is, focal lengths. The blue darkroom has a small focal length and records a large part of the subject (in blue). The red darkroom has a large focal length and records a small part of the subject (in red).

Notice that because the image sensors of the two darkrooms are of the same size, the red darkroom with large focal length has a smaller angle of view and can record more details of the visible subject part thanks to a larger magnification. On the contrary, the blue darkroom with smaller focal length has a larger angle of view but can record less details of the subject because of the smaller magnification.

When choosing a lens, its focal length determines which part of the subject will be captured (as determined by the angle of view). This part of the subject recorded by the camera is called the “[frame](#)”. This frame is visible in a

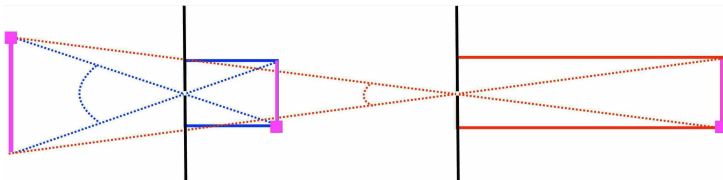
rectangle in the viewfinder or directly on the screen (see section 12).

Spatial telescopes have huge focal lengths such as 57.6 meters (189 ft) for the [Hubble Space Telescope](#) and 131.4 meters (431 ft) for the [James Webb Space Telescope \(JWST\)](#). Photo lenses have much shorter focal lengths, typically between 10 mm and 800 mm,

10 Most Common Focal Lengths of Lenses

The most common [Leica M lenses](#) have focal lengths of 28, 35, or 50 mm. The 28 mm, with large angle of view (approximately 75 degrees horizontally), is typically used for a landscape, the 35 mm with a smaller angle of view (54 degrees) is better fitted for a group of persons, while the 50 mm with even smaller angle of view (47 degrees) will be used for an individual (although, obviously, these lenses can be used in all circumstances but produce different photos).

To take a picture of a subject of a given size with a lens of smaller angle of view, one can get farther from the subject. Symmetrically, to take a picture of a subject of a given size with a lens of larger angle of view, one can get closer from the subject.



In both cases the subject will be captured with exactly the same size on the image sensor.

But the pictures will be different! For example¹, The



Figure 3: Perspective distortion

background appears farther for lenses of small focal lengths and closer with lenses of larger focal lengths, the 50 mm being very similar to the human eye. Notice also that the green glass² in the background is much more blurry with the 120 mm that it is with the 28 mm. The explanation is given in section 22 (depth of field).

11 Perspective Distortion

The backgrounds in the pictures of figure 3 look quite different because of the different [perspective distortions](#), that is, what is in front and behind the subject is different on the three pictures.

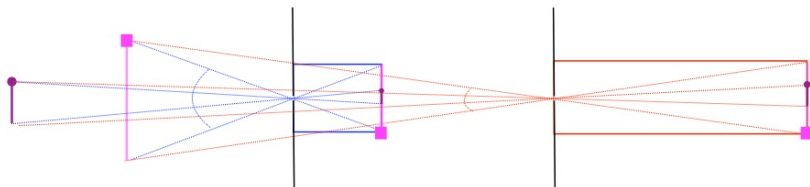
For example with a 28 mm, what is behind the subject looks very far since it will be small. With 50 mm what is

¹The pictures have been taken with the [Leica LUX app](#) on an [iPhone](#), there is no Leica 120 mm M-mount lens.

²[Couleur menthe à l'eau](#), [Eddy Mitchell](#).

behind the subject will be larger and look similar to what we see with human eyes. With the 120 mm focal length, the background is larger so looks closer.

This perspective distortion becomes clear on the following schema.



The subject (in magenta) appears to be of the same size on the image sensor of both darkrooms. But the purple object in a distance behind the subject is smaller on the image sensor of the blue camera with short focal length and larger on the image sensor of the red camera with longer focal length.

The further is an object, the smaller it looks like. Therefore, in the picture of the blue camera the brown object will look farther (since it is smaller) and it will look closer on the red camera (since it is larger).

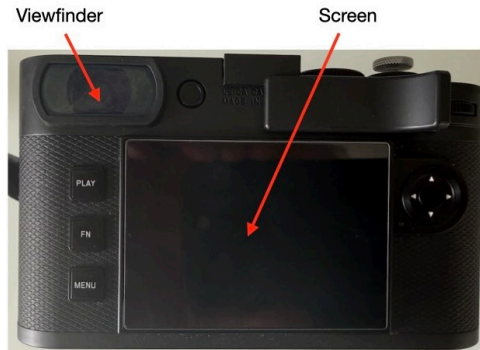
It follows that the three 28 mm, 35 mm, and 50 mm lenses can all be used to take pictures of the subject at a given size (provided enough space is available for the photographer to move farther or closer to the subject) but, unless there is no distant background, the three pictures will look quite different.

12 Viewfinder and Screen

In darkrooms the inverted picture appears directly on the translucent sensor. In Leica M cameras, the picture (in fact the inverted image straighten) can be seen through the viewfinder or on the screen (called LCD panel) for recent cameras (since the M8 but not for the M11-D which has no screen).

At the time of [film photography](#), the photographer could not immediately see the result and had to wait a few days until [development](#) of the [photographic film](#) in an amateur [darkroom](#) or a [specialized commercial laboratory](#).

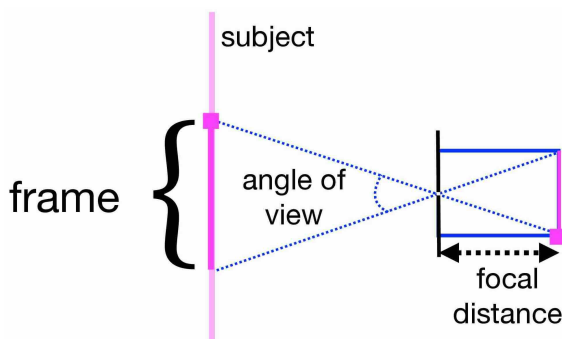
The [viewfinder](#) and [LCD liquid crystal screen](#) both allow the photographer to anticipate what will appear in the pictures taken by the camera.



(the camera is off on this picture). The viewfinder and screen of the M EV1 both show the subject as seen through the lens (plus other informations to be discussed later).

13 Frame

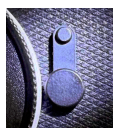
The **frame** is the part of the subject reflected on the electronic image sensor (and therefore the viewfinder and screen of the M EV1). Only this frame part of the subject will appear on the photo.



The frame depends on the focal length of the lens. This frame appears exactly and identically on the screen and in the viewfinder.

It is possible to select electronically part of the frame to get a zoom effect.

In front of the camera, there is a frame lever.



Frame selector lever

Pushing the selector lever away from the objective (that is to the left when behind the camera), a rectangle appears on the screen, to zoom $\times 1.3$ on the center of the frame.



1.3 zoom

Pushing again the selector lever away from the objective, shows a smaller rectangle zooming $\times 1.8$ on the center of the frame.



1.8 zoom

Notice that this is an electronic zoom, selecting part of the photographic sensor, so that the photo resolution is reduced. A lens adapter allows for optical zoom (see section 46).

14 Camera

A [camera](#) is a darkroom with a image sensor capturing light like a metal plaque covered by silver salts darkening with light (which yields a [negative](#)). The darkroom image sensor was originally prepared by the photographer in the dark and then covered to be protected from light. The pinhole of the darkroom is closed, the image sensor is introduced in the back of the darkroom and uncovered. It is now sensitive to light. The photographer then opens the pinhole for long

enough for the image sensor to capture enough light. This time is called the “**exposure time**”. In the early days of photography it was hours, later seconds, and nowadays can be fractions of milliseconds.

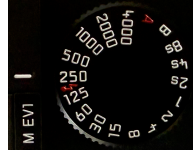
The first **Leica cameras** used **photographic films**, nowadays **electronic image sensors**.



15 Exposure Time (or Shutter Speed)

Originally, the exposure time was very long, a few hours, since the image sensor was not very sensitive to light. Photography was for immobile subjects only, such as landscapes. A person traversing the landscape would not appear on the photo since it did not produce a large enough quantity of light to impress the image sensor. Over time, sensitivity of image sensors improved and it became possible to take pictures of persons, provided they did not move for a few dozen of seconds. Photographers used supports of the head and body to prevent movements so that the photographed persons often looked tense, rigid, and cramped!

On the Leica M EV1, the exposure time (or shutter speed) can be chosen thanks to the shutter speed dial on top right of the camera.



The shutter speed dial can be turned left or right and set to be 1/4000 s ([second](#)), 1/2000 s, 1/1000 s, 1/500 s, 1/250 s, a red lightning ⚡ (1/180 s) for photos taken with a flash, 1/125 s, 1/60 s, 1/30 s, 1/15 s, 1/8 s, 1/4 s, 1/2 s, 1 s, 2s, 4s, or 8s, each time doubling the exposure time. There is also a B meaning that the exposure takes place as long as the shutter button remains pressed down. Finally the A means that the exposure time will be automatically chosen by the camera image sensor and software.

16 Over/Correct/Under Exposure

The exposure measurement is visible on the exposure scale at the bottom of the screen or that of the viewfinder

If a photo is overexposed, that is, too bright, with too much light captured, this will be indicated by a positive number on the exposure scale in the viewfinder and on the screen



is overexposed

(the speed is too slow).

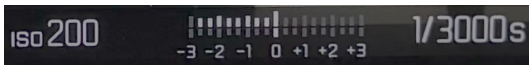
A solution is to increase the speed, by turning the shutter speed dial left.

Else, if a photo is well-exposed, the exposure scale in the viewfinder and on the screen will be on zero.



is well exposed

Otherwise, the photo is underexposed, that is, too dark, with not enough light captured, as indicated by negative number on the exposure scale.



is underexposed

(the speed is too fast)

A solution is to increase the exposure time, by turning the shutter speed dial right which will decrease the speed.

17 Camera Stand

However, beyond $1/250$ s to $1/60$ s, the photographer will possibly slightly move the camera and the photo will be blurry. A solution is to use a [monopod](#), a [tripod](#) or minipod.



tripod

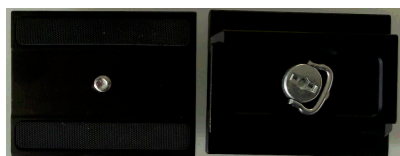


minipod

The Leica minipod can be screwed directly on the bottom of the camera.



An adapter may be needed for tripods.



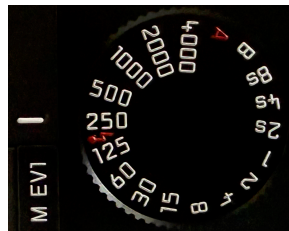
A mechanical shutter release cable may be used to avoid any movement of the camera on the tripod when pressing the trigger.



The shutter release cable has a wheel to block it in down position for the B long exposure time.

18 Shutter Speed

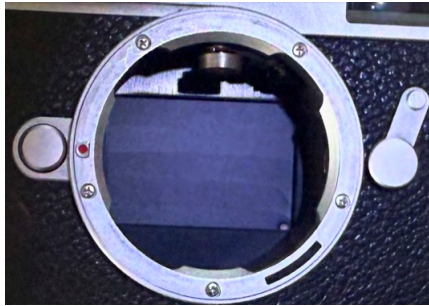
The shutter is the physical device on the camera that opens and closes to control the exposure time of the electronic sensor as determined by the shutter speed dial on top right of the camera (that can be moved by half increments).



The faster (respectively slower) is the shutter speed the smaller (resp. larger) is the exposure time so less (resp. more) light touches the camera electronic sensor.

The exposure time is a time measured in seconds. Calling it **shutter speed** is somewhat a confusing misunderstanding, since a **speed** is measured in **meters per second** (or fractions of these). But obviously, the faster the shutter moves (in m/s), the shorter the exposure time (in s). Calling the shutter speed the exposure time avoids the confusion.

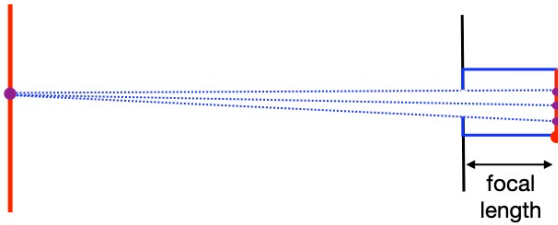
The shutter can be seen in closed position on a camera with no lense.



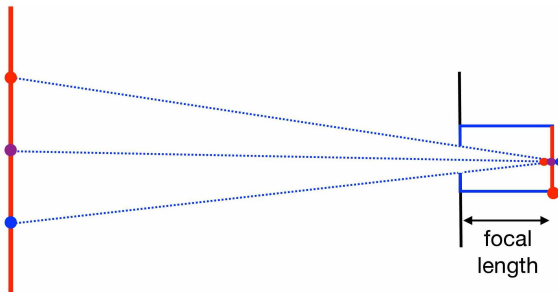
The mechanical shutter consists of light metal blades moving very quickly (in up to $1/4000$ s). The shutter is very fragile and should not be touched or blown on (with one's breath or a rubber dust blower ball). The M EV1 has also an electronic shutter to control the light exposure of the sensor electronically allowing exposure times up to $1/16000$ s. Electronic shutter are not perfect so that the M EV1 offers, by default, an hybrid mode, using the mechanical shutter up to $1/4000$ s exposure time and the electronic shutter beyond.

19 Lenses

An obvious solution for a darkroom to capture more light is to have a larger hole. But then a point of the subject will send light rays to different points of the image sensor.

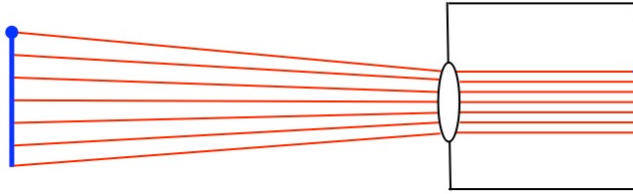


Symmetrically, a point of the image sensor will receive light from different points of the subject.

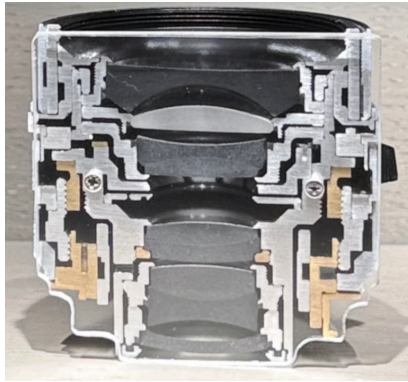


The result is that the photo will be blurry.

To allow a larger hole in a darkroom without blurring the image sensor, photographers invented [lenses](#) (that they also call [objectives](#) or glasses in slang). Ideally, a lens would project exactly the subject in reduced size on the image sensor.



In practice it is impossible to achieve this ideal goal [39, section 4.3.1, page 50] and lenses have a much more **complicated design**, always with some limitations [16].



(courtesy [Leica Store, New York](#))

There are usually several lenses in the lens³ and some mechanism to move them within the lens for focussing, as well as a second mechanism, called a diaphragm, to determine the aperture to modify the size of the hole through which the light beams reach the image sensor, so called entrance pupil (as well as a third related one mechanically transmitting the lens aperture to the camera computer).

³In french the lenses inside the lens are “lentilles” inside the “objectif” without ambiguity. **Photographic objective** is rarely used by photographers in English.

20 Aperture

Lenses have an adjustable [iris diaphragm](#) allowing for different [apertures](#) with large holes (called [entrance pupil](#)) having small numbers while small holes/entrance pupils have large numbers. Typical apertures are $f / 0.95$, $f / 1.4$, $f / 2$, $f / 2.8$, $f / 4$, $f / 5.6$, $f / 8$, $f / 11$, $f / 16$, $f / 22$. Small f -numbers (also called focal ratio, f -ratio, or f -stop) correspond to large apertures through which a large quantity of light goes through while large f -numbers correspond to small apertures through which a small quantity of light goes through. The desired aperture is chosen by turning the focus ring marked with these f -numbers. Looking through the lens clearly shows the various apertures (which exact size depends on the lens).

A lens with a small f -number is called [fast](#) because when wide opened it captures a lot of light so that the exposure time can be small (or shutter speed very fast). The fastest Leica lens is the NOCTILUX-M 1:f/0.95 50mm ASPH (which first appeared in 2008).

21 Focussing

Ideally whatever is the distance of the subject to the lens, the subject should appear sharp on the photo. Unfortunately, this is not possible in practice. Lenses must be focussed on the subject for this subject to appear clear, sharp, in focus on the photo.

On Leica M cameras, [focussing](#) is [manual](#). A particular case is when the subject is far enough (usually more than 15/20 m), in which case the distance is set to ∞ (infinity,

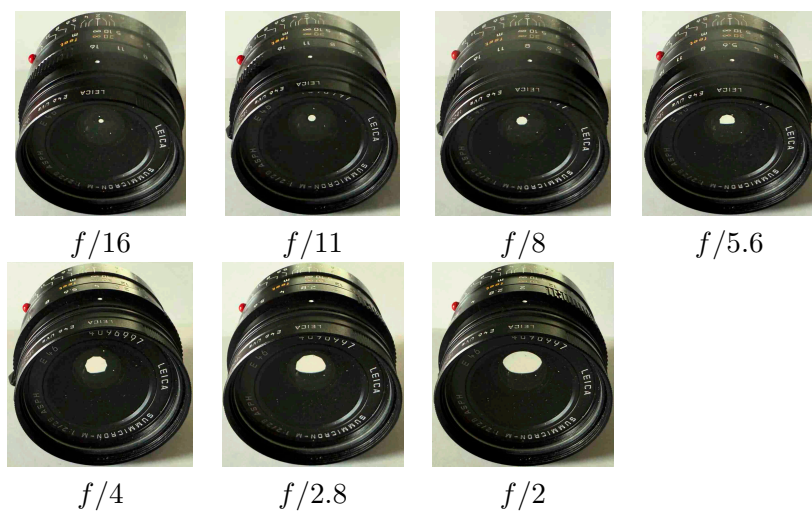


Figure 4: Entrance pupil of the SUMMICRON-M 1:2/28 ASPH lens with UV filter and no hood nor rear cap at apertures $f / 16$, $f / 11$, $f / 8$, $f / 5.6$, $f / 4$, $f / 2.8$ and $f / 2$. Half increments are also possible but not shown.

math for very far!).



Taking pictures of landscapes is always easy since focussing is trivially set to ∞ ! Otherwise, the photographer must be set manually to the distance to the subject.

The [focussing distance](#) in photography is measured from the camera's electronic sensor plane to the subject, not from the front of the lens. This sensor plane is about $1/3$ in the back of the M EV1 camera.

21.1 Focussing by Measuring ([Optional](#))

The focussing distance can be measured by a [tape measure](#) or a [laser distance meter or rangefinder](#) (0.947 m in our example) and the focus ring turned to be positioned at that distance (1 m in the example), with some tolerance since the measure is often more precise than necessary.



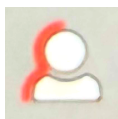
Although most photographers do not use this measuring method, it is very precise when extremely [sharp image quality](#) is required.

21.2 Focussing with the Viewfinder or the Screen

It is possible to focus with the viewfinder or the screen, turning slowly, left or right, the focussing ring until the image is clear on the screen. Three settings of the camera do help.

21.2.1 Setting Focus Peaking On

Pushing the frame lever towards the objective (to the right) will set focus peaking on and off. The following icon (visible on the left of the viewfinder and the right of the screen)



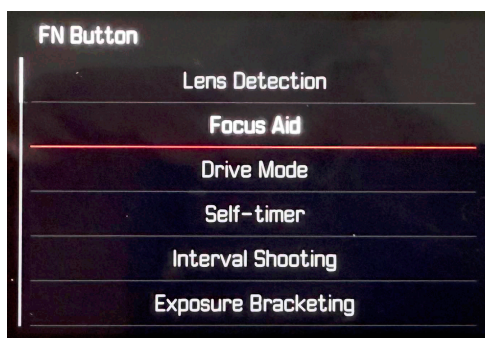
means that focus peaking is on, which helps for focussing. If this symbol does not appear, focus peaking is off, and focussing can only be done by looking at the sharpness of the picture in the viewfinder or on the screen.

21.2.2 Magnification

- Pressing down, long enough, the function button, on the top right of the camera,



a menu appears on the screen



Using up or down arrows of the directional pad, select **Focus Aid** and then press the **Center** button to select this option.

- Afterwards, shortly pressing this function button will magnify the middle of the screen, which is helpful for focussing. Pressing the function button again will go back to the full image on the screen or viewfinder.

21.2.3 Focus peaking

When on , focus peaking highlights the edges of in focus subject elements in red.



On the left screen the parts of the plant in focus have a red border. On the right screen no part appears in red since the camera focus is incorrect.

22 Depth of Field

For each of its possible apertures and focus distances, a lens has a corresponding **depth of field**, that is a zone where the photo is clear, sharp, in focus. Subjects outside that this field will appear blurry on the photo (see figure 5). Of course the transition from sharp to blurry is progressive. The closer in front or farther behind the subject, the more blurry the photo will be. Photographers appreciate a smooth transition. Depending on the lens and aperture, the depth of field can be very large or tiny. Each lens has a

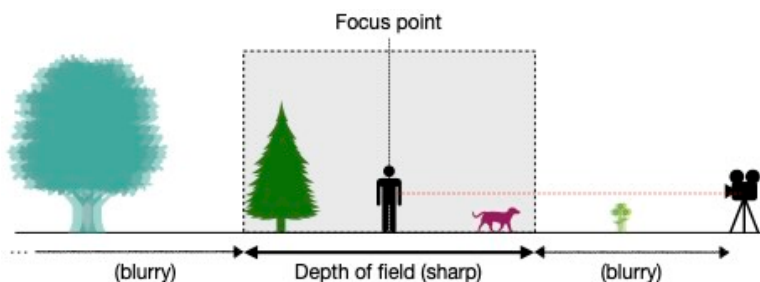


Figure 5: Depth of field

different depth of field which depends on the aperture and is engraved on the unmovable depth of field ring of the lens.



Depth of field ring of the SUMMILUX-M 1:1.4/35 ASPH

Lenses with small focal length have the largest depth of field. Here is the depth of field of the SUPER-ELMAR-M 1:3.8/18 ASPH with focal length of 18mm.



As shown on the middle picture, the aperture is set to $f/8$ and the focus distance to 5 m (15 feet for the [British Imperial System](#)). The depth of field starts at 1.1 m and is sharp up to infinity ∞ (and beyond :). If the aperture is set to 16, then the depth of field for a focus distance of 5m goes from 0.7 m (on the left picture) to ∞ (and beyond on the right picture). This means that, with this lens, choosing an aperture of 16 and a focus distance of 5 m, all pictures will be in focus.

Lenses with large focal length have the smallest depth of field. Here is the depth of field of the APO-TELYT-M 1:3.4/135 of focal length 135mm with focus distance of 5 m.



For the smallest aperture of 22, the depth of field has a minimum of 4.3 m and a maximum of 6.3 m, which means that the sharp zone, in gray in figure 5, is only two meters deep. At the maximal aperture of 3.4 the depth of field is very small, a few [decimeters](#). Nevertheless this lens is no problem for photographing objects at a long distance such as the summit of a mountain in a landscape or the top of a [skyscraper](#).

Small depths of field have been used in [portraiture](#). For example with an APO-SUMMICRON-M 1:2/90 lens of focal length 90mm at maximum aperture of 2 and a distance to the subject of 1 m



the depth of field is tiny (and not much larger with aperture 16). Therefore, the subject will be sharp but its background blurry, the farther, the blurriest.

In summary,

depth of field:	small	large
aperture:	large (small f -number)	small (large f -number)
focal length:	large	small

A shallow depth of field is in general suitable for a close subject while a greater depth of field may be necessary for a distant subject.

23 Hyperfocal distance (hyperfocus)

The [hyperfocal distance](#) (or [hyperfocus](#)) is the focussing distance that maximizes the depth of field for a given lens and aperture. When focussing at the hyperfocal distance, the depth of field extends from half that distance to infinity.

M EV1 (which, moreover allows for thousands of photos on SD cards). The purely mechanical [Leica M-A](#) camera and the [Leica M6](#) incorporating an electronic [light meter](#) are the most popular recent Leica film cameras.

25 ISO (Sensitivity)

The films had different [sensitivities](#) (improperly called “[film speed](#)”)⁴, the more sensitive films requiring a smaller quantity of light, therefore allowing for faster shutter speeds.

A standardized method of [sensitometry](#) was introduced in 1934 and [internationalized](#) in the 1974 by the [ISO \(International Organization for Standardization\)](#) to [measure film speed](#).

Popular film ISOs are 100, 200 and 400 but one can also find films at ISO 25, 50, 64, 160, 800, 1600 and 3200. A doubling of film sensitivity is represented by a doubling of the numerical film speed value.

A small ISO produces the best quality photos while the quality may degrade at high ISOs with the appearance of [noise](#), such as unwanted [grain](#), dots, and lines.

When the ISO is automatic (by MENU → ① → ISO → Auto ISO), the photographic sensor ISO is automatically chosen by the camera and can be upper limited to avoid noise (by MENU → ② → Auto ISO Settings → Maximum ISO → ISO 3200, for example).

Otherwise the ISO can be chosen by the photographer by MENU → ① → ISO → ISO 3200, for example.

The base ISO of the M EV1 is 60. This is the lowest ISO setting on the M EV1, which provides the highest image

⁴The obsession with “speed” dates back to the origin of photography where exposure times were extremely long.

quality with the least amount of noise. The traditional film sensitivities of 100, 200, 400, even 800 are also popular. Beyond 3200/4000 ISO the quality may slightly degrade.

26 Automatic Mode

As discussed in section 3, the **Camera Settings** can be chosen automatically by the M EV1, Just select the ISO **Auto ISO**, the shutter speed **A**, the aperture according to the ambient luminosity and desired depth of field (for example 5.6 inside and 16 outside), and select the subject distance manually using the viewfinder or screen, knowing that ∞ will always work for landscapes. The M EV1 will automatically select the ISO and shutter speed. The [algorithm](#) used by the M EV1 for this selection is sketched in section 33.4, optionally, for interested readers only.

27 Manual Settings of the Camera

To take a photo, the manual settings of the M EV1 are

1. **Lens:** The choice of the lens, which determines the frame, that is the part of the subject appearing on the photo, and the magnification.

Moving closer or further from the subject with any lens may be possible, but this will affect the perspective, as discussed in section 11;

2. **ISO:** The sensitivity of the electronic sensor measured in ISOs. The lower the better to get better photo qual-

ity (64 offering the best quality, 400 is often chosen as equivalent to a good quality film sensitivity).

3. **Aperture:** The aperture of the lens, the smaller the better (that is, the larger f -number the better) to allow for a deep depth of field (unless bokeh is desired, see section 40);
4. **Shutter speed:** For the shutter speed, the faster is the better to avoid blurring with hand held cameras (but blurred photos by moving the camera or choosing low shutter speed is also popular to get off the beaten track [35, page 76]);
5. **Focussing:** Finally the manual focussing on the subject.

One can start with a standard choice like (ISO 400, aperture $f / 5.6$ or $f / 8$ inside and $f / 16$ or $f / 22$ outside, and a shutter speed of $1/250$). Then one can either use the information in the viewfinder or screen to adjust the speed as explained in section 31 or take the picture, and adjust these settings empirically from the photo on the screen (this is one reason for photographers to look at the screen after shooting).

Another solution is to take a first picture in automatic mode, look at the setting chosen by the camera, adjust if necessary, and retake the photo with the adjusted settings. Of course this will work well for [architectural photography](#) but probably not for [street photography](#).

Taking several shots is recommended since failed pictures cost nothing and are easy to erase by `PLAY → MENU → Delete Single → FN` (marked with a trash), moving to other

pictures with $\triangleleft$ and $\triangleright$ arrows, and terminating with PLAY (marked $\leftrightarrow$).

The M EV1 can also be instructed to take several pictures (3 or 5) for a shot at different exposures and speeds. See *exposure bracketing* in the manual [23, page 108].

28 Histogram

When pressing the shutter bottom halfway, the M EV1 displays a preview of the photo together with other informations in transparent superposition, such as an [image histogram](#)⁵.

The image histogram represents the number of [pixels](#) in the image for each color brightness in the picture, darker colors on the left and lighter colors on the right.

- The image details may be lost due to blown-out highlights. For example, the sky may look white with no cloud visible. In this case, the histogram has a peak on the right;
- On the opposite, the image details may be lost due to blacked-out shadows. For example the shadow of a building looks black with nothing visible within the shadow. In this case, the histogram has a peak on the left;

Both phenomena appear on the following picture so that the histogram (appearing superimposed on the screen preview) has two peaks,

⁵If there is no histogram set MENU $\rightarrow$ ④ $\rightarrow$ Capture Assistants $\rightarrow$ Info Profiles $\rightarrow$ Info Profile 1 $\rightarrow$ and set Histogram to On. The Grids and Level Gauge (to check the camera horizontality and verticality) may also be useful.

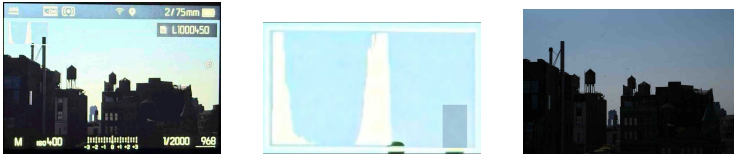


Photo histograms may have different shapes but peaks and biases to the left or right always show an excess in some color brightness, which, in general, is undesirable.

29 Stops: Aperture versus Exposure Time versus ISO

If, for a given amount of light, a photo is too bright/overexposed (respectively dark/underexposed), the photographer must decrease (respectively increase) the **exposure**, that is the quantity of light received by the camera electronic sensor. There are three possibilities (often, somewhat improperly, called the “exposure triangle”, except that a triangle has three sides).

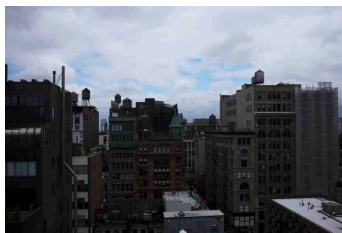
- **Aperture:** decrease (respectively increase) the aperture (that is increase (respectively decrease) the f -number);
- **Shutter speed:** decrease (respectively increase) the exposure time, that is, increase (respectively decrease) the shutter speed;
- **ISO:** The ISO modification was listed last since it was not available with film photography. This fixed ISO was a limitation which disappeared with digital cameras. **ISO:** decrease (respectively increase) the ISO of the image sensor.

The increment or decrement in these cases are called “[f-stops](#)” or simply “stops” by photographers. For example [stopping down](#) goes down by one stop (for example from 11 to 8) while stopping up goes up by one stop (for example going up from 4 to 5.6).

Most lenses allow to stop the aperture up or down by a half-stop. Some lenses like the [Carl Zeiss Distagon 2.8/15 ZM](#) even offer 1/3 stops (that is 19 possibilities between 2.6 and 22).

Cameras and lenses are designed so that an aperture stop, a shutter speed stop, and an ISO stop allow for the same change of the quantity of light received (up by doubling it, down by dividing it by 2). For math fans, this will be explained in detail in optional section [33](#).

Assume we have used the lens [SUMMILUX 1:1.4/35 ASPH](#) to take the following picture



16/250/200

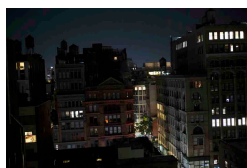
at aperture 16, speed 250, and ISO 200. If we think that is too dark (a question of taste), we can make the following corrections (on top is the camera screen and below the photo).



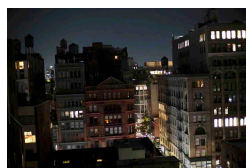
The photos look pretty the same since in each case the change was by 1 stop.

So photographers have to look for a compromise since increasing the aperture (decreasing the f -number) may restrict the depth of field, decreasing the speed may yield blurring, and increasing the ISO may introduce noise.

Moreover the results of the camera and lens settings depend on the lens. Figure 6 is an example of two photos taken during the night with the SUMMILUX-M 1:1.4/50 ASPH and the NOCTILUX-M 1:0.95/50 ASPH lenses, both at aperture $f/1.4$, speed 1/30 s, and ISO 3200 focussed at ∞ . Although the



SUMMILUX-M 1:1.4/50 ASPH



NOCTILUX-M 1:0.95/50 ASPH

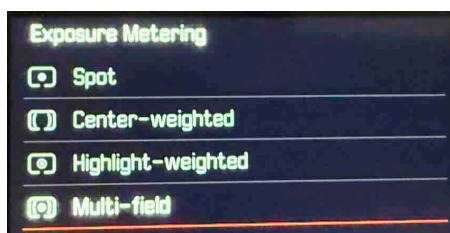
Figure 6: Different brightness for same settings
 settings are the same, the NOCTILUX-M 1:0.95/50 ASPH photo looks

brighter. The lens brightness depends on its [design](#) (number and type of elements), [coating](#), and handling of optical aberrations (like [chromatic](#) and [spherical aberrations](#)) but also on its “speed” (minimal aperture f -number), see sections [33.10](#) and [33.11](#).

30 Exposure Metering With the Camera

To decide automatically of the correct camera settings for a photo, the M EV1 determines the [exposure](#), that is the quantity of light per unit area reaching the surface of the [electronic image sensor](#) (and then showing its estimation on the screen or viewfinder to help the photographer get the correct exposure as previously explained in section [27](#)).

The M EV1 lets the user choose how the exposure is computed, by MENU → ① → Exposure Metering showing the following alternatives



- The **Multi-field** default option determines the exposure by looking at the whole frame;
- The **Center-weighted** option determines the exposure by looking around the middle of the frame;

- The **Highlight-weighted** option determines the exposure by prioritizing the bright areas (to prevent them to be overexposed and appear white);
- The **Spot** option determines the exposure by looking at a small disk in the middle of the frame.

This may be useful for example to take a picture of the moon in the dark. With the **Multi-field** option, most of the picture is dark, which requires a high sensitivity, so that the moon will be very over exposed. But with the **Spot** option pointing at the moon, its exposure will be correct, while the rest will remain dark, more precisely under exposed dark, which does not make much difference.

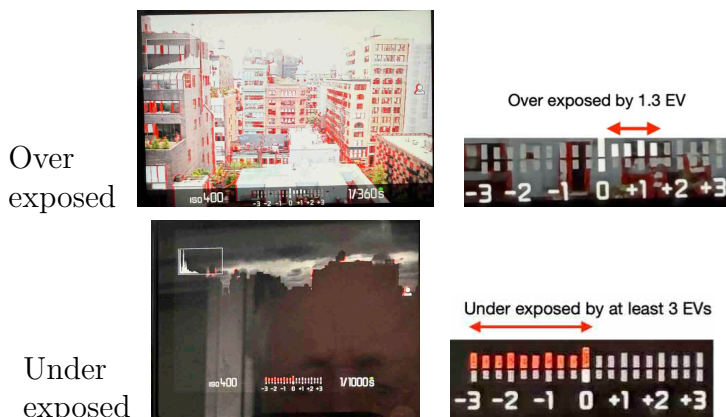
31 EVs (Exposure Values)

An **EV (Exposure Value)** is a measure of the quantity of light that must be captured by the camera sensor (film or electronic), as defined by given ISO, apertures, and exposure time, according to the light brightness reflected by the subject. The stops as discussed in the previous section 29 correspond to 1 EV.

To indicate whether a photo is overexposed (too much light captured by the electronic image sensor) or underexposed (not enough light captured by the electronic image sensor)

A scale in EVs is shown in the viewfinder and on the screen. This scale shows the *difference* between the expected brilliance according to the camera settings (ISO, aperture, and shutter speed) and the actual brilliance of the subject

(as measured by the image sensor), both expressed in EVs, that is stops.



When the exposure is correct, the scale is on 0 EV



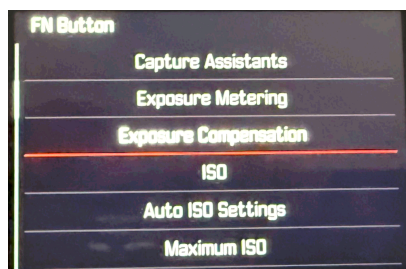
EVs are further explained in next section [33](#) (for the math nerds only).

32 EV Compensation (or Correction)

In automatic mode the aperture is chosen by the photographer (to get an appropriate depth of field) and the exposure (shutter speed, and ISO) is chosen by the camera software. The camera choice may not be that desired by the photographer, being too dark or too bright. An instruction can be given to the M EV1 to go brighter (by a positive number of (thirds of) EVs) or darker (by a negative number of (thirds

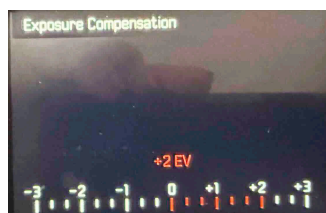
of) EVs). Since this correction may depend on each photo taken, it is useful to make this correction easily accessible, as follows:

- Press down the toothed thumbwheel (on top right of the back of the camera), a menu will appear



and select the Exposure Compensation to assign the exposure compensation selection to the thumbwheel.

- Afterwards, pressing the thumbwheel and turning it left or right *while taking a picture* will select an exposure compensation measured in thirds of EVs.



This will be added to the exposure selected by the camera or photographer (hence subtracted, that is darker, for negative values).

33 A (Long) Digression on the EV Formula for Math Lovers (*Optional*)

If you are resistant to mathematics, just skip this section 33 and go directly to the next section 34, with [one click](#), with a light heart and no regret. A definitely more ambitious solution would be to first refresh your knowledge in mathematics by reading [5].

33.1 The EV Formula

But if you like formulas, here is one

$$EV = \log_2 \left(\frac{100 \times f^2}{I \times s} \right) \text{ where } \begin{array}{l} EV : \text{Exposure Value} \\ I : \text{ISO} \\ f : \text{lens aperture} \\ s : \text{shutter speed} \\ \text{in seconds} \end{array} \quad (1)$$

An EV is a number representing the ISO I , aperture f , and exposure time (or shutter speed) s to be set on a camera for the camera sensor (film or electronic) to capture a given quantity of light during s seconds. Lower EV numbers indicate low light (so more light is needed for the photo) while higher EVs indicate more light (so less light is needed for the photo). This means that an EV is also a measurement of the subject [brightness](#) based on camera settings (different from but related to [luminance](#) based on [photometry](#) in optics). In a dark, unlit room, the exposure time with ISO 100, during 1 s, with aperture of $f/1.0$ of the camera is $EV = \log_2 1 = 0$ (see figure 7). If the quantity of light captured by the camera sensor and the subject brightness coincide, the photo is well-exposed.



In a dark, unlit room Pointing at lit light bulb

Figure 7: EV measurement

All camera combinations of the ISO, f -number, and shutter speed that yield the same exposure have the same EV. An incrementation of the EV by 1 corresponds to 1 stop (either on the ISO, aperture, or shutter speed), as explained in section 29 and proved in section 33.17.

As shown in figure 7, a subject brightness EV can be measured by an external [light meter](#) (see section 48 for more details).



For example an exposure $EV = 14$ can be achieved by $ISO = 100$, $f = 22$, and $t = \frac{1}{30}$ since $\log_2(22^2 \times 30) = 13.8257538329 \simeq 14$ (among other solutions).

An external exposure value measurement is not necessary since the Leica M EV1 can measure the EV of the light reflected by the subject with its image sensor. Given a measured EV , formula (1) provides the possible ISO I , aperture f , and shutter speed s , that can be chosen to have the correct exposure for that measured EV .

Notice that the camera can read the exposure time thanks to the position of the shutter speed dial (see section 15), the ISO thanks to the camera setting (see section 25), and the aperture f -number thanks to a mechanical connection of the interchangeable lenses and the camera described in section 33.2.

33.2 Communication of the Lens f -number to the Camera

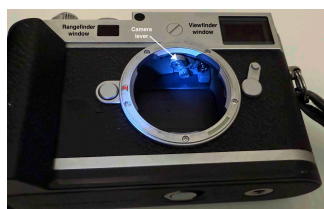
The mechanical coupling of the focussing ring to the pentagonal prism of the rangefinder in the camera is the same for all M-lenses of any focal lens. So it is part of the lens design to move the camera lever (on the right image of figure 8) the same way for all lenses at the same focussing distance. You



control cam
position at 0.7m



control cam
position at ∞



camera lever

Figure 8: Mechanical link between the lens focus and camera

can compare for example figure 8 for the SUMMILUX-M 1:1.4/35 ASPH lens with figure 9 for the much smaller SUMMARON-M 1:5.6/28 lens. The position of the control cam visible through



Figure 9: Notch and a control cam on a lens

the notch is the same at the same focus distance despite the fact that the two objectives have different sizes hence need different extensions of the lens for focussing.

This also means that you can use your camera to measure the distance to a (not too close) subject by focussing on it, with any lens.

33.3 Use of the EV Formula in Manual Mode

In manual mode, the Leica M EV1 measures the *EV* of the light reflected by the subject with its image sensor (depending on the Multi-Field, Center-Weighted, and Spot metering mode, see section 30), reads the *f*-number, ISO, and shutter speed that the photographer has chosen on the lens and camera, computes the right-hand side of formula (1), and compares it with the *EV* measured with its image sensor. The result of this comparison is shown with a 1/3 EV graded scale on the the viewfinder and screen. For example



means underexposed by 2 EVs.

33.4 Use of the EV Formula in Automatic Mode

In Auto ISO and shutter speed automatic mode **A**, the Leica M EV1 measures the *EV* of the light reflected by the subject with its image sensor (depending on the metering mode), get the *f*-number from the lens aperture, and uses formula (1) to determine the ISO and shutter speed (which are shown on the screen, the speed only in the viewfinder). The balance between ISO and shutter speed is influenced by MENU → ② → Auto ISO Settings allowing to set up limits on the ISO and shutter speed

- The Maximum ISO (for example ISO 4000), and
- The slower Shutter Speed Limit (for example $1/(4f)$ s (which is a rule of thumb for setting the minimum shutter speed to avoid motion blur caused by handheld camera shake, a bit more tolerant than the traditional “1/focal length” rule) or $1/60$ s (a simplification of the old $s \leq 1/f$ second rule of thumb to avoid blurring for focal length $f/50$ which isn’t on the speed dial, so the next fastest option of $1/60$ is chosen).

Starting from the base ISO 64 for the M EV1 and maximum speed ($1/4000$), the camera evaluates formula (1) and terminate with these settings in case of equality with the measured *EV*. Otherwise, because of low light, the camera will successively increase the shutter speed by one stop ($1/4000 \rightarrow 1/2000 \rightarrow 1/1000 \rightarrow 1/500 \rightarrow \dots$) until a solution is found or the Shutter Speed Limit is reached. Then the ISO is increased by one stop, and the procedure is repeated.

If no solution is found with maximal ISO and minimal speed, the camera software will try harder, by bypassing the user settings and going to the camera limits. But if the photo is ultimately too dark (for example because you forgot the lens cap) you get  and a black photo. The only solution is to remove the cap or to illuminate the subject. If the photo is too bright, you get  and the photo is white. A solution is to use a ND filter (see section 42) and move to the electronic shutter allowing higher shutter speeds by MENU → ③ → Shutter Type → Electronic.

This mode of operation allows the photographers to use the formula without having to know it, since, given the selected aperture f , the ISO and shutter speed are selected automatically to get the right settings, if any. The settings will chose the lowest possible ISO and fastest possible shutter speed.

33.5 Use of the EV Formula in ISO Manual Mode

The ISO can be chosen by the photographer at ISO values between 64 and 50000 by MENU → ① → ISO → ISO n where n is the ISO value chosen between 64 and 50000. Assuming that the shutter speed is set to A, the shutter speed will automatically determined using formula (1) for the chosen ISO n value.

33.6 Apertures of a Lens

To understand formula (1), let us first understand the magic aperture numbers 1.4, 2, 2.8, 4, 5.6, 8, 11, 16, 22 appearing

on the aperture ring of lenses, already discussed in section 20.

The powers of the square root of 2 are [5, page 122]

- $\sqrt{2}^1 = \sqrt{2} = 2^{\frac{1}{2}} = 1.41421356 \simeq 1.4$ and $1.41421356 \times 1.41421356 = 1.99999999 \simeq 2$ (by $\sqrt{a} \times \sqrt{a} = a^{\frac{1}{2}} \times a^{\frac{1}{2}} = a^{\frac{1}{2}+\frac{1}{2}} = a^1 = a$ since $\sqrt{a} = a^{\frac{1}{2}}$ and $a^m \times a^n = a^{m+n}$);
- $\sqrt{2}^2 = (2^{\frac{1}{2}})^2 = 2^{\frac{1}{2} \times 2} = 2^1 = 2$ (by $((a^m)^n) = a^{mn}$);
- $\sqrt{2}^3 = 2.82842712 \simeq 2.8$;
- $\sqrt{2}^4 = \sqrt{2}^2 \times \sqrt{2}^2 = 2 \times 2 = 4$;
- $\sqrt{2}^5 = 5.65685425 \simeq 5.6$;
- $\sqrt{2}^6 = 8$;
- $\sqrt{2}^7 = 11.3137085 \simeq 11$;
- $\sqrt{2}^8 = 16$;
- $\sqrt{2}^9 = 22.627417 \simeq 22$.

So the apertures engraved on the lens are the rounded powers of $\sqrt{2}$.

33.7 Effect of the Aperture Diameter on the Quantity of Light Captured by the Sensor Per Second

Lenses have an aperture ring marked by different f -numbers, such as 1.4, 2, 2.8, 4, 5.6, 8, 11 or 16 for the SUMMILUX-M 1:1.4/50

ASPH. The lens has a mechanism, called a [diaphragm](#), that will change the diameter d of the aperture (also called [entrance pupil](#)) when turning the aperture ring to different f -numbers (see figure 4). This changes the size of the entrance pupil hence the quantity of light reaching the image sensor during a shot (say of 1 second).

The entrance pupil is a disk (or very close to be a disk). Therefore, the surface of the entrance pupil is πr^2 where r is the radius of the entrance pupil set by the diaphragm when turning the aperture ring, that is $\pi(\frac{d}{2})^2 = \pi\frac{d^2}{4}$ for the diameter $d = 2r$.

Increasing aperture by one stop, will decrease the diameter by a factor $\sqrt{2}$ so the surface of the aperture disk is now $\pi(\frac{d}{2 \times \sqrt{2}})^2 = \pi\frac{d^2}{(2 \times \sqrt{2})^2} = \pi\frac{d^2}{2^2 \times \sqrt{2}^2} = \pi\frac{d^2}{4 \times \sqrt{2}^2} = \pi\frac{d^2}{8} = \frac{1}{2}\pi\frac{d^2}{4}$. Therefore increasing the aperture by one stop divides the surface $\pi\frac{d^2}{4}$ of the aperture by 2, hence divides by 2 the quantity of light received per second.

Decreasing aperture by one stop, will increase the diameter by a factor $\sqrt{2}$ so the surface of the aperture disk is now $\pi(\frac{d \times \sqrt{2}}{2})^2 = \pi\frac{(d \times \sqrt{2})^2}{2^2} = \pi\frac{(d^2 \times \sqrt{2}^2)}{4} = \pi\frac{(d^2 \times 2)}{4} = 2 \times \pi\frac{(d^2)}{4}$. Therefore decreasing the aperture by one stop multiplies the surface of the aperture by 2, that is, doubles the quantity of light received per second.

We remember from this section that the quantity of light received by the image sensor from a lens with entrance pupil diameter d is proportional to $\pi\frac{d^2}{4}$ that is of the form $c_a d^2$ where c_a is a constant coefficient that depends only on the lens/objective (for example the quality of the [glass](#) and [coating](#) of the lenses in the objective). Because a stop changes d by a factor $\sqrt{2}$, $c_a d^2$ is changed by a factor 2.

33.8 Effect of the Focal Length on the Quantity of Light Captured by the Sensor Per Second

The [inverse square law of distance in optics](#) is a [law of physics](#), that is a mathematical formulation of an immutable observation of a natural phenomenon. This inverse square law of distance states that the intensity of light from a point source is inversely proportional to the square of the distance from the source. For example doubling the distance reduces the quantity of light received to one quarter. This is clearly visible when using a flash, since close subjects are much brighter than distant ones on the photos.

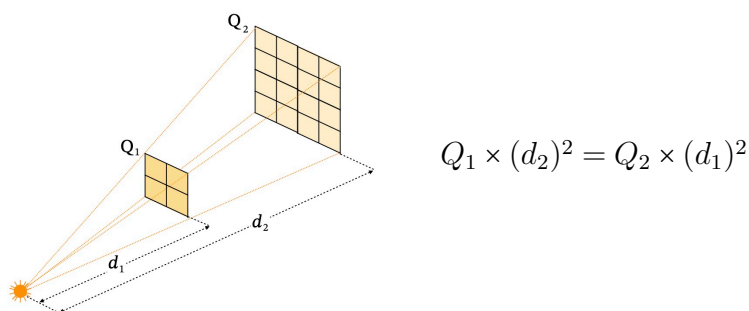


Figure 10: Inverse square law of distance in optics

If a subject reflects the same quantity of light through lenses of different focal lengths fl_1 and fl_2 with the same aperture (that is same entrance pupil diameter d), this quantity of light will be more dispersed on a larger part of the sensor with large focal lengths and more concentrated on a smaller part of the sensor with small focal lengths, as illustrated on figure 10. Considering that the light source is the

entrance pupil, we have

$$\frac{Q_1}{(f_l)_2^2} = \frac{Q_2}{(f_l)_1^2}$$

We remember from this section that the quantity of light received per second by the image sensor from a lens with a focal distance fl is inversely proportional to fl^2 , that is of the form $c_{fl} \frac{1}{fl^2}$ where c_{fl} is a constant coefficient that depends only on the lens/objective (for example the [optical clarity](#) of the lenses in the objective).

33.9 Combined Effect of the Aperture Diameter and the Focal Length on the Quantity of Light Captured by the Sensor Per Second

We can now combine the effect of the aperture diameter d of the lens and the chosen focal lens fl on the quantity of light reaching the image sensor per second. It increases as $c_a d^2$ with the diameter d but is attenuated by a factor $c_{fl} \frac{1}{fl^2}$ because of spreading the gathered light over an area on the sensor inversely proportional to the square of the focal distance fl . The quantity of light receives per second by the sensor is of the form $c_a d^2 \times c_{fl} \frac{1}{fl^2}$ that is $c \frac{d^2}{fl^2}$ where $c = c_a \times c_{fl}$ is a constant specific to the lens.

33.10 Entrance Pupil Diameter

Lenses (of focal length fl) are always designed to have a diaphragm, such that, for each f -number f chosen on their aperture ring, the entrance pupil is opened to a diameter d

satisfying

$$f = \frac{fl}{d} \quad \text{or} \quad d = \frac{fl}{f} \quad (2)$$

This design ensures that any two lenses with different focal lengths fl_1 and fl_2 but same aperture f receive the same quantity of light per second (or very close ones).

To prove it, consider two lenses of respective focal lengths fl_1 and fl_1 set on the same aperture with f -number f . Then the quantity of light each lens receives per second is, by section 33.9,

$$\begin{aligned} & c_i \frac{(d_i)^2}{(fl_i)^2} && \text{for lens } i = 1, 2 \\ = & c_i \frac{(\frac{fl_i}{f})^2}{(fl_i)^2} && \text{by (2)} \\ = & c_i \frac{(\frac{fl_i}{f})^2}{(fl_i)^2} && \text{by } (\frac{a}{b})^n = \frac{a^n}{b^n} \\ = & c_i \frac{(fl_i)^2}{f^2 \times (fl_i)^2} && \text{by } \frac{\frac{a}{b}}{c} = \frac{a}{b} \times \frac{1}{c} = \frac{a}{b \times c} \\ = & \frac{c_i}{f^2} && \text{by simplification } \frac{a}{a} = 1 \text{ and } a \times 1 = a. \end{aligned}$$

This calculation proves that the two lenses receive the same quantity of light per second since in practice $c_1 \simeq c_2$ although they might be slightly different, as shown in figure 6 at the end of section 29, because of slightly different lens designs or use of different [UV filters](#) capturing more or less visible light.

33.11 Speed (Maximal Aperture) of a Lens

The “speed” of a lens is its minimal f -number F , corresponding to its maximal aperture. The “speed” is an inappropriate term, but follows from the fact that fast lenses

(such as $F = 0.95$ or $F = 1.4$) allows for rapid shutter speeds, that is small exposure times, resulting in less blurring.

The “speed” of a Leica lens is written on that lens, for example, the SUMMILUX-M 1:1.4/50 ASPH is a lens of focal length 50 mm and maximal f -number $F = 1.4$. In general it will be written $1 : F/fl$ where F is the maximal aperture of the length ($F = 1.4$ in fact $\sqrt{2} = 1.41421356$ in our example) and fl is the focal length ($fl = 50$ mm in our example).

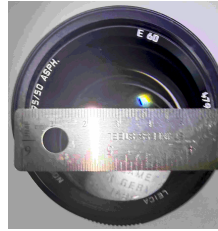
Applying equation (2) to this maximal aperture case, we get the maximal aperture F available which corresponds on that lens to a diameter d_m , such that

$$F = \frac{fl}{d_m} \quad \text{or} \quad d_m = \frac{fl}{F} \quad (3)$$

So for the SUMMILUX-M 1:1.4/50 ASPH the diameter d_m will be $\frac{50}{1.4} = 35.7714$ while for the NOCTILUX-M 1:0.95/50 ASPH is will be $\frac{50}{0.95} = 52.631$, as shown on the following pictures



1:1.4/50



1:0.95/50

The $1 : F/fl$ designation on Leica M lenses derives from $d_m = \frac{fl}{F}$ in equation (3). The “1:” prefix in Leica M lens designations signifies a division that is $\frac{1}{F}$ while $/$ is multiplication by fl since $1 : F/fl = (\frac{1}{F})/fl = \frac{1}{F} \cdot \frac{1}{fl} = \frac{1}{F \cdot fl}$. For example

“1:2/75” on the APO-SUMMICRON-M 1:2/75 ASPH lens represents a maximum aperture diameter $d_m = \frac{1}{\frac{2}{75}} = \frac{75}{2} = 37.5$ mm⁶. Equivalently, a lens of focal lens 75 mm and maximal aperture 37.5 mm has a “speed”, maximal aperture, or minimal f -number 2.

33.12 Effect of the ISO on the Quantity of Light Captured by the Sensor Per Second

The ISO [film speed](#) (more precisely film sensitivity nowadays electronic image sensor sensitivity) has been defined so that when doubling the ISO, the film sensitivity doubles, so that the quantity of light fixed on the film or image sensor per second doubles. Taking the base ISO 100, the ISO 200 doubles the quantity of light captured per second, ISO 400 quadruples it, and so on for ISO 800, 1600, 3200, 6400, etc. So the contribution of the ISO sensitivity I to the quantity of light captures by the sensor per second is $\frac{I}{100}$. Moreover doubling the ISO doubles the quantity of light captures per second, that is, corresponds to one stop.

33.13 Effect of the Exposure Time (Shutter Speed) on the Quantity of Light Captured by the Sensor Per Second

The shutter speeds on the dial of figure [3](#), are $\frac{1}{4000}$, $\frac{1}{2000}$, $\frac{1}{1000}$, ..., $\frac{1}{125}$, $\frac{1}{60}$, ..., $\frac{1}{15}$, $\frac{1}{8}$, $\frac{1}{4}$, $\frac{1}{2}$, 1, 2, 4, 8, being each time multiplied by 2 (or so for $\frac{1}{125} \rightarrow \frac{1}{60}$ and $\frac{1}{15} \rightarrow \frac{1}{8}$ as well as

⁶The solidus symbol / was introduced by the English logician [Augustus de Morgan](#) in 1845 while the colon : was introduced by the German mathematician [Gottfried Leibnitz](#) in 1659.

$\frac{1}{n}$ written n to ease engraving of shutter speed on the dial, but the camera software that reads the dial positions can use the exact shutter speed values).

The quantity of light captured by the image sensor is proportional to the exposure time (shutter speed) s . This is the quantity received during one second multiplied by s so is of the form $c_s s$ where c_s is the quantity of light captured during one second. It follows that changing the shutter speed by one stop down double the quantity of light captures per second, that is, corresponds to one stop.

33.14 Combined Effect of the Aperture, Focal Length, and ISO on the Quantity of Light Captured by the Sensor Per Fraction of Second

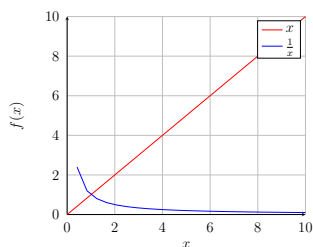
We have seen that the contribution of the lens focal length fl and aperture (f -number) f is $\frac{c}{f^2}$ (thanks to the clever choice (2) of the diameter d of the entrance pupil). That of the ISO I is $\frac{I}{100}$. That of the shutter speed s is $c_s s$. Combining these effects is multiplicative, since one stop down on the ISO and one stop down on the aperture and one stop down on the speed divide the quantity of light received by the sensor by $\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}$. It follows that the cumulated effect of the aperture, focal length, and ISO on the quantity of light captured by the sensor during s seconds is $\frac{c \times I \times c_s s}{f^2 \times 100}$ of the form $q \frac{I \times s}{100 \times f^2}$ where q is the quantity of light captured by the image sensor at ISO 100, for an f -number 1, during 1 second. The coefficient q can be measured precisely by [photometry](#). The formula $q \frac{I \times s}{100 \times f^2}$ is close to (1) but different because we have reasoned on the quantity of light received by the film or image sensor per second whereas (1) is about

the exposure value.

33.15 What is the EV (Exposure Value)

The quantity of light $q \frac{I \times s}{100 \times f^2}$ captured in s seconds increases when increasing the ISO I , the exposure time s , or the aperture (which decreases the f -number) so that the photo will be brighter, if not overexposed. So higher values of $q \frac{I \times s}{100 \times f^2}$ indicate more light captured.

On the contrary, EVs are not about the quantity of light received but about the camera settings to cope with a given quantity of light received. So if the quantity of light received is very high, we must decrease the ISO, shutter speed, and aperture (that is increase the f -number) to capture less of the received light so that higher EVs indicate less light to capture. If the quantity of light received is very low, we must increase the ISO/aperture/speed (that is decrease the f -number) to capture more of the received light so that lower EV numbers indicate more light to capture. This corresponds to the inverse $\frac{1}{q \frac{I \times s}{100 \times f^2}} = \frac{100 \times f^2}{q \times I \times s}$, as shown when comparing x (in red) and $\frac{1}{x}$ (in blue),

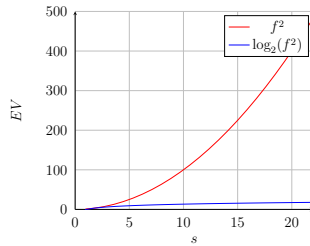


Decreasing the ISO I or the shutter speed s or decreasing the aperture by increasing the f -number will increase the quantity $\frac{100 \times f^2}{q \times I \times s}$, that is increase the quantity of light reaching the

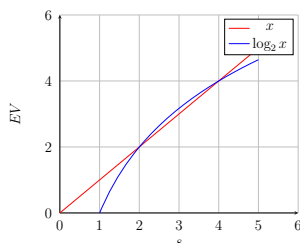
image sensor. If more light reaches the image sensor, it must capture less of the light for a correct exposure. Therefore, like the EV, higher values of $\frac{100 \times f^2}{q \times I \times s}$ indicate less light to be captured while lower values of $\frac{100 \times f^2}{q \times I \times s}$ indicate more light.

33.16 Why the logarithm?

Without the [logarithm](#) $\log_2$ [5, page 443] in formula (1), the value $\frac{100 \times f^2}{q \times I \times s}$ grows rapidly. For example, with $q = 1$, ISO $I = 100$, and shutter speed = 1 s, the EV, which is f^2 in this case, is plotted in red as a function of f and rapidly reaches large numbers. By applying the $\log_2$ the numbers remains small (say less than 25, bright sun on snow being about 17). This is clearly shown by $\log_2 f^2$ plotted in blue. Moreover multiplications become additions (since $\log_2 a \times b = \log_2 a + \log_2 b$) which simplifies calculations based on $\sqrt{2}$ (since $\log_2(\sqrt{2}) = \frac{1}{2}$ and $\log_2(2) = 1$).



Because units can be chosen arbitrarily (such as the [Kelvin](#) (K), [Celsius](#) (°C), and [Fahrenheit](#) (°F) scales for [temperature](#)), we can arbitrarily scale by $\log_2$ in (1) that is $\log_2(q \frac{I \times s}{100 \times f^2})$ = $\log_2(\frac{I \times s}{100 \times f^2}) + \log_2 q$ and ignore the translation term $\log_2 q$. Moreover the logarithm is an increasing function. The larger is x , the larger is $\log_2 x$, as shown thereafter.



It follows that the requirement that higher EVs indicate less light to be captured while lower EVs indicate more light, as satisfied by $\frac{100 \times f^2}{I \times s}$, is preserved by taking the logarithm $\log_2$.

This idea of **logarithmic scale** is frequent in physics to measure quantities with a broad range of values such as **exponential growth**. Another example is **decibels** to measure **sound intensity levels** relative to a reference value.

Finally, we have (hopefully) explained the meaning of the *EV* formula (1).

33.17 EVs and Stops

Observe that to stop up by one stop, we can either

- Increase the aperture by 1 stop, which multiplies f by $\sqrt{2}$ so that $(f \times \sqrt{2})^2 = 2 \times f^2$ so that the logarithm $\log_2$ increases the EV by 1 (since $\log_2(2x) = \log_2 x + \log_2 2 = \log_2 x + 1$), or
- Increase the ISO by 1 stop, that is double it, so that, again, the logarithm $\log_2$ increases the EV by 1, or finally
- Decrease the shutter speed by 1 stop, which multiplies s by 2, so that, once again, the logarithm $\log_2$ increases the EV by 1.

We conclude that stopping up by one stop increases the *EV* by 1, and similarly stopping down by one stop decreases the *EV* by 1.

More details on the [exposure value](#) can be found on Wikipedia, in particular its relation with the physical notion of [luminance](#) to characterize the [brightness](#) of subjects.

It just remains to cite the German [camera shutter](#) manufacturer [Friedrich Deckel](#) who is credited with developing the Exposure Value (EV) system in the 1950s.

34 Leica M-mount Lenses Designation

Back to pragmatism, the Leica M-mount lenses are designated by esoteric [names](#), often of [latin](#) origin, depending on the maximal aperture of the lens:

- [Noctilux](#): maximal apertures of $f / 0.95$ or $f / 1.0$ or $f / 1.2$ or $f / 1.25$;
- [Summilux](#): maximal apertures of $f / 1.4$, $f / 1.5$ or occasionally $f / 1.7$;
- [Summicron](#): maximal apertures of $f / 2$;
- [Elmarit](#): maximum aperture of $f / 2.8$;
- [Summaron](#): maximum aperture of $f / 2.8$ or $f / 3.5$ or $f / 5.6$;
- [Elmar](#): maximum aperture of $f / 3.8$ or $f / 4$ (Tri-Elmar is for a lens offering three different focal lengths);

This name can be preceded by

- APO: [apochromatic lens](#) enforcing all colors to focus on the sensor at the same distances from a lens;

and the designation can be followed by

- ASPH: [aspherical lens](#) to reduce optical aberrations.

The name is followed by $1:f/n.n$ where $f/n.n$ is the maximum aperture (with minimal f -number)⁷ and then the focal length (in millimeters mm).

Lenses are also marked *Edd*, where *dd* is a number indicating the diameter in millimeters of the filters that can be screwed in front of the lens. For example, the APO-SUMMICRON-M 1:2/90 ASPH lens is marked E55.

Other terms and abbreviations related to Leica lenses are explained on the [Understanding Leica Lenses](#) page on Leica web site.

35 Lens Mark

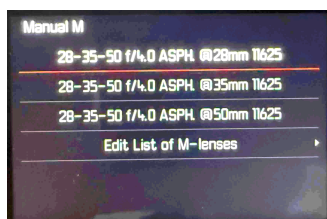
Recent Leica lenses have marks on their mount to distinguish them and transmit this focal length and maximum aperture information to the camera to optimize image quality, see figure 11. The information can also be shown on viewfinder and the screen, used by the camera software to correct lens defects, and encoded in the digital photos. Choose the default MENU → ① → LensDetection → Auto for marked lenses. Unmarked Leica lenses can be selected manually by MENU → ① → LensDetection → Manual M then select the lens and On.

⁷The "1:" prefix in Leica M lens designations signifies a division. For example, "1:2/50" represents a ratio indicating the relationship between the lens's focal length and its maximum aperture diameter. So, "1:2/50" signifies a 50mm lens with a maximum aperture diameter that is half of its focal length ($50\text{mm} / 2 = 25\text{mm}$). This translates to a maximal aperture of $f/2$. More details in section 33.7.



Figure 11: The white and black [binary](#) mark of the APO-SUMMICRON-M 1:2/75 ASPH lens on the left and the unmarked TRI-ELMAR-M 1:4/28-35-50 ASPH lens on the right.

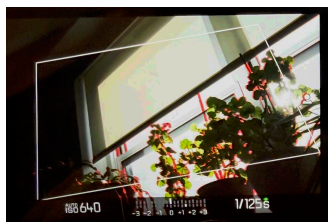
For the TRI-ELMAR-M 1:4/28-35-50 ASPH lens at 28mm this is



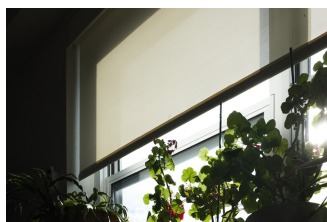
Choose MENU → ① → Lens Detection → Off for non-Leica lenses (or use some similar Leica lens instead, if any).

36 Perspective Control

Perspective control allows the position of the camera to be taken into account to correct the perspective. The [perspective projection](#) (where parallel lines converge at infinity on the photo) is replaced, for the part of the picture visible in a rectangle, by the more natural [parallel projection](#) (where parallel lines remain parallel on the photo).



screen



photo

Perspective control is selected by MENU → ③ → Perspective Control → On.

Since perspective control may not be desirable for all pictures, it can be assigned to the function button FN. Press the function button FN long enough, select **Perspective Control** with the down arrow of the directional pad and **Center** button.

37 Profile

The camera settings (called a profile) can be saved in the camera memory and reused each time the camera is turned on.

37.1 Setting a Profile

A simple profile is the following.

- ON → MENU → ⑤ → Reset Camera → Yes (and No to all other questions). Restarting the camera, choose the **Language**, the **Time Zone**, **time**, **Daylight Saving Time**, **Date Format**, and **date**.
- Press down the small function button on top-right of the camera



- and choose **Focus Aid** in the menu. Then, this button can be used for precise focusing as explained in section 21.2.
- Assign **Exposure Compensation** to the thumbwheel as explained in section 32.
- Assign **Perspective Control** to the function button FN as explained in section 36.
- Limit the maximum ISO as explained in section 25.
- Optionally, show the histogram, grid, and **level gauge** on the screen as explained in section 28.
- Choose the format and size of digital photos by MENU → ② → File Format → JPG. The DNG format is necessary only for say 2x3 meters photos or when **cropping** part of the photo (see section 13). Then select **JPG Settings** → Max JPG Resolution → S-JPG (18 MP) is enough for most family photos.
- To save these settings, MENU → ④ → User Profile → Manage profiles (well below) → Save as profile → Users 1 → Yes. If the camera is turned OFF and then ON, these settings for User1 will be automatically used.

37.2 Using a Profile

To use profile **User1**, select MENU → ④ → User Profile → User1 → **Active**. This profile will then remain active when the camera is turned Off and On. Before each photo, it remains to use

either the automatic mode in section 26 or the manual mode of section 27.

38 Lens Flare

Lens flare and **glare** is often caused by very bright sources, giving the impression that the photo has been invaded by too much light. The problem is generally solved by slightly reorienting the camera.



Modern lenses are **coated** to avoid this phenomenon as much as possible by altering the way in which the lens reflects and transmits light. A **lens hood** can also reduce lens flare.

39 Vignetting

The subject seen in a disk by a lens must be projected to the 2:3 rectangle sensor, which may have undesirable optical effects. One of them is **vignetting** that is a reduction of an image's brightness on its periphery, often in the corners. Here is a first photo of a white wall taken with the

Carl Zeiss DISTAGON 2.8/16 ZM lens, where vignetting is visible. By screwing a filter which is darker in the center in front of the lens (Carl Zeiss Center Filter (-1,5 EV) for the Distagon 2,8/15 ZN), the vignetting is attenuated, on the second photo, at the expense of exposure, which is reduced by the filter.



40 Bokeh

Bokeh is the effect described in section 22, where the depth of field is very small so that the subject is sharp with proper focussing whereas the foreground and background are blurry. Leica M 50mm, 75mm, and 90mm lenses are generally appreciated in portraiture for their smooth bokeh. The bokeh can be accentuated by a **center spot filter** to further blur the image periphery.

In this example, taken with a handheld M-EV1 equipped with the 2025 re-issue of the SUMMILUX-M 1:1.4/50 CLASSIC $f/1.4$ lens at 0.7 m and 1/200 with close focussing with the screen on the yellow **spadix** of a **spathe** of *Anthurium*, the foreground and background are blurry. This lens is known for its smooth creamy bokeh.



The bokeh can be accentuated using a [warm center spot filter](#).

41 UV Filters

A [UV filter](#) (or UV ([ultraviolet](#)) pass filter) is often screwed in front of the lens to protect the glass and its [coating](#). This may harm [contrast](#) and [sharpness](#) and be at the origin of lens flare (see section [38](#)). An alternative is to use a [hood](#) adapted to the lens.

42 ND Filters

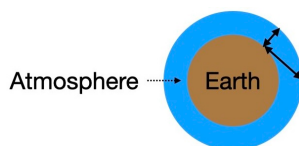
[ND \(neutral-density\) filters](#) can be used to reduce the quantity of light entering the lens (without changes in color rendition), for example to extend the exposure time. Variable neutral-density filters have several positions allowing to modify the quantity of light blocked.



An exposure correction using the ISO, shutter speed, and aperture is usually preferable, except for exceptional situations (such as long exposures of very bright subjects that would yield overexposed photos or to get bokeh in a portrait with a large aperture and bright lightning which cannot be compensated by a low ISO and fast speed).

43 Rayleigh Scattering

Everywhere on earth, the atmosphere is thinner when looking overhead and thicker when looking towards the horizon.



It follows that the blue sun light is less filtered by the atmosphere when looking straight up than it is when looking towards the horizon. This is called [Rayleigh scattering](#). The result is that the sky is a much darker blue at the top of a photo than it is at the horizon, where it can even be white. The eye and the brain compensate but the camera doesn't. Moreover the dynamic range of the eye is much better than that of a camera, which means that a camera cannot capture details both in bright and darker regions of the subject. There are three possible remedies.

- use a neutral-density filter, see section [44](#);
- use a polarizing filter, see section [45](#) and [Rayleigh sky model](#);
- extend the camera dynamic range by shooting at 36MP or 18MP (MENU → ② → DNG Resolution → S-DNG (18 MP)) or, in JPEG mode, choosing a high extended dynamic range (MENU → ② → JPG Settings → Extended Dynamic Range → High).

44 ND Graduated Filters

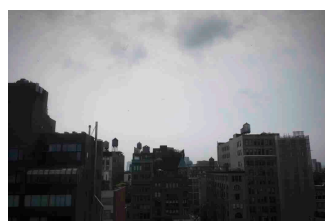
A [graduated neutral-density \(ND\) filter](#) can be used to reduce the quantity of light entering part of the lens, typically half of it, with a gradual transition from one half to the other.



It can be used to darken a bright sky so that both the sky and subject can be properly exposed.



without ND Grad filter



with ND Grad filter

45 Circular Polarizer

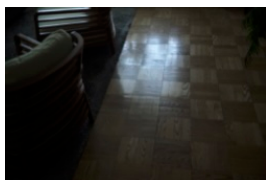
A [circular polarizer](#) can be mounted in front of the lens to reduce [light polarization](#) (for the 16-18-21mm lens below).



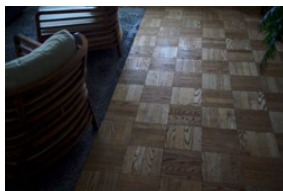
A [filter](#) adapter may be necessary to adjust to the lens size.

By slowly turning the mobile part of the circular polarizer right or left, polarization will be attenuated (but not

if the sun is in front or just behind) or even completely suppressed (if the sun is at 90 degrees).



Polarized light



Reduction
of polarization
with a circular filter

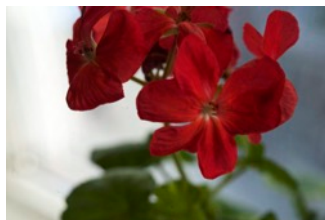
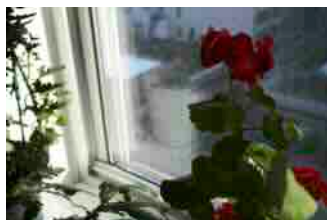
46 Macro Photography

The M EV1 offers two possibilities for [Macro photography](#):

- The magnifying glass Leica ELPRO E52 can be screwed in front of some lenses (in the following example a Voigtländer NORTON 40mm F 1.2)



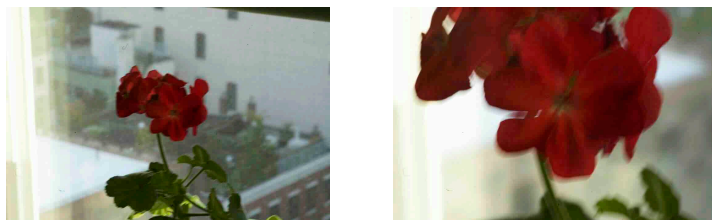
The FN (function) button can be used to magnify the viewfinder and screen view of the subject ⁸. The magnifying effect of the Leica ELPRO E52 is seen by comparing the following two pictures



- The Leica Macro-Adapter-M, here mounted on a Leica MACRO-ELMAR-M 1:4/90 lens (which must be extended by turning and pulling the front of the lens before using) extends the focal length of M-lenses, which has a magnifying effect, as explained in section 11.



⁸This is set up by maintaining the FN (function) button pressed down until a menu appears and selecting Focus Aid.



Notice that zooming in section 13 or computer applications such as [Darktable](#), [GIMP](#), [GraphicConverter](#), [Lightroom](#), [Luminar Neo](#), or [Preview](#) on [MacOS](#) and their counterparts on [Linux](#) and [Windows](#) can be used to select part of a image, but this cropping reduces the [image resolution](#), which is not the case with macro photography.

47 Cleaning the Electronic Sensor

If dust reaches the electronic sensor it will appear as dots on photos. The sensor can be cleaned but this is a very delicate operation better done by a specialist at a Leica store, as found at [URL \(universal resource locator\) leica-camera.com](https://www.leica-camera.com) or the [Leica Camera official service page](#).

48 Using an Exposure Meter for Correct Exposure (*Optional*)

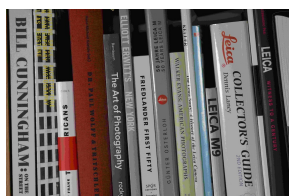
A [light meter \(or illuminometer\)](#) can measure the amount of light. In an exposure meter, the light meter is coupled to either a [digital](#) or [analog calculator](#) which displays the correct ISO, shutter speed, and f -number for optimum exposure. For example one or two value are chosen by the photographer and the other(s) is/are calculated.

48.1 Camera Measurement of the Reflected Light

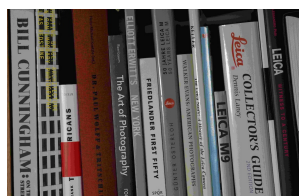
The exposure is measured by the camera (as explained in section 30) using its image sensor. This measures the [light reflected](#) by the subject. The camera has a computer to determine the correct ISO and shutter speed for a given lens at a given aperture (f -number).

These exposure measurement are used by the M EV1 in automatic mode (ISO, speed, or both) to choose the camera settings.

For example, we take a photo of books with an M EV1 on a tripod with an APO-TELYT-M 1:3.4/135 lens at $f/8$ and 1.7 m with different exposure measures. With automatic ISO and speed, the automatic choice with multi-field exposure metering is ISO 2500 and 1/320 s while spot metering (on the middle white book) yields ISO 2000 and 1/250 s.



$f/8$, ISO 2500, 1/320 s
multifield

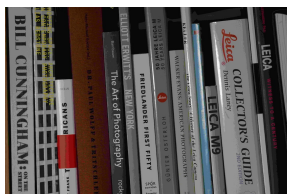


$f/8$, ISO 2000, 1/250 s
spot metering

In manual mode the M EV1 uses the exposure measurement to inform the photographer about the validity of his own settings, see section 31.

For example at 400 ISO, the manual choice of the correct speed is 1/60 s, as indicated by 0 EV in the viewfinder or screen





$f/8$, ISO 400, $1/60$ s

48.2 Exposure Meter Measurement of the Incident Light with a Handheld Exposure Meter

A handheld exposure meter is an alternative method to measure the amount of light.

These exposure meters can be placed just in front of the subject to measure the incident light directly received by the subject from a source (like the sun or an artificial light). Since the amount of light is measured directly from the source, the measurement is usually more precise than the amount of light from the source reflected by the subject (which may [absorb](#) some of it).

These exposure meters have a white bulb (called lumisphere) through which the received light quantity is measured. (They can also measure the reflected light, usually when used without the bulb).

48.2.1 Spherical incident light measurement

When the bulb is out/on, the incident light is diffused through the bulb, so that the measure is the mean of the light received around the bulb (that is by the subject on which the instrument is placed).



$f/8$, ISO 400, $1/24$ s

48.2.2 Flat incident light measurement

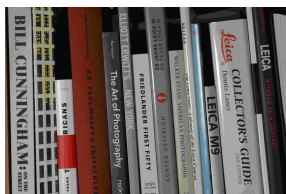
When the bulb is in/off, the measure is restricted to the light amount received exactly at the bulb point.



$f/8$, ISO 400, $1/15$ s

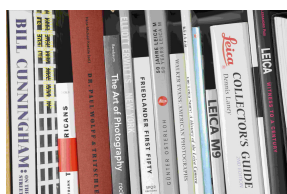
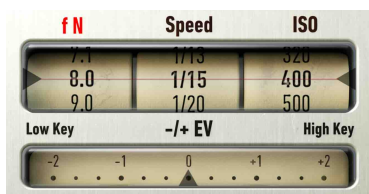
48.3 Exposure Meter Measurement of the Reflected Light

Exposure meters can also be used to measure the light reflected by the subject (so called reflected light measurement). In that case the meter is placed near the camera and pointed to the subject. The zone measured is a disk which size is determined by a fixed or variable angle degree (for example 1 to 40 degrees, 20 degrees in our example) from the meter cell.



$f/8$, ISO 400, $1/50$ s

Phones have light meter applications operating through the phone lenses that are generally less precise than dedicated instruments.

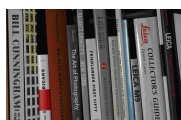


$f/8$, ISO 400, $1/15$ s

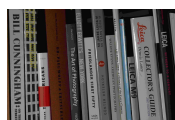
Notice that the previous examples have slow speed but the image is not blurred since the camera is on a tripod. All examples look pretty the same although they are not.



ISO 2500, $1/320$ s



ISO 2000, $1/250$ s



ISO 400, $1/60$ s



ISO 400, $1/50$ s



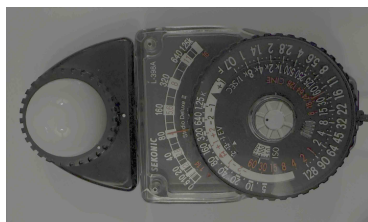
ISO 400, $1/24$ s



ISO 400, $1/15$ s

Handheld exposure meters can be coupled to a [flash](#) to measure the light received when flashing.

An exposure meter may be indispensable for purely mechanical film cameras, with a rangefinder but no electronics at all, such as the [Leica M-A \(Typ 127\)](#). [Selenium meters](#) were purely mechanical and existed well before the electronic age, so work without battery. They use a [selenium photocell](#) to produce electricity from light and moving a needle on the meter. Most modern versions use [amorphous silicon photodetector](#) to measure the intensity of illumination on a surface.



(the measure of 80 [foot-candles](#) (about 7.43 [lux](#)) at ISO 100 is manually reported under $\hat{H}$ so that the proper exposure is 1/250 s at $f/5.6$, 1/30 s at $f/16$, etc.)

49 Conclusion

We have explained the basic concepts in photography and elementary use of the Leica M EV1. We have added optional technical explanations to appreciate the ingenuity of photographers over almost two centuries!

The [Leica M EV1 instruction manual](#) [23] is indispensable to go beyond an elementary use of the Leica M EV1 and explore its numerous other possibilities.

Research on photography goes on, in particular in computer science, where it is called [computational photography](#). The research results are published in scientific conferences such as the annual [IEEE International Conference on Computational Photography](#),

Numerous books are available to explain the historical [21, 30, 31, 42], technical [2, 9, 24, 35], and artistic aspects [1, 3, 4, 6, 7, 8, 10, 11, 12, 13, 14, 15, 17, 18, 20, 22, 25, 26, 27, 28, 29, 32, 33, 34, 37, 38, 40, 41] of photography, including for Leica cameras [19, 21, 30, 31, 35, 36].

Magazines, like [LFI](#), [Modern Photography](#) and [Popular Photography](#) ([PopPhoto](#)), review contemporary photography.

Online discussion groups such as [Irys](#) can be used to share interest in photography.

Going to *photography museums* (such as the [International Center of Photography](#) (ICP), [The Met](#), or the [MOMA](#) in New York, the [Photography centre](#) in London, the [Albert Kahn museum](#) in Boulogne-Billancourt near Paris, the [Maison européenne de la photographie](#), the [Fondation Henri Cartier-Bresson](#), the [Centre Pompidou](#) in Paris, or the [Maison de la photographie Robert Doisneau](#) in Gentilly near Paris), visiting *temporary exhibitions* (in particular in [Leica stores](#)) and *photography galleries* (like [Danziger](#) and [Howard Greenberg](#) in New York, [Autograph](#) and [The Photographers' Gallery](#) in London, [Polka](#) and [Le Bal](#), or the avant-garde [François Bourdoncle](#), in Paris), attending *photography festivals* (such as [Photoville](#) in Brooklyn, [Paris Photo New York](#) in New York, NY, USA, [FORMAT](#) in Derby, [Photo Oxford](#) in Oxford, [Photo London](#) in London, UK, and the [Rencontres d'Arles](#), the [Visa pour l'image](#) in Perpignan, and the [Paris Photo](#), France), as well as joining *photography collectives* (such as [NYC Street Photography Collective](#) or the [London Alternative Photography Collective](#)) can also be an inexhaustible source of endless inspiration.

Happy [photo shooting](#)! (More precisely, [focussing](#), [fram-](#)

ing, and shooting :))

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Introduction to Photography with the Leica M EV1

Patrick Cousot

A short, simple, and illustrated introduction to the fundamental concepts of photography (with a few optional technical explanations), and their practical application with a Leica M EV1.

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